Inspiration Seeds: Learning Non-Literal Visual Combinations for Generative Exploration

ANONYMOUS AUTHORS

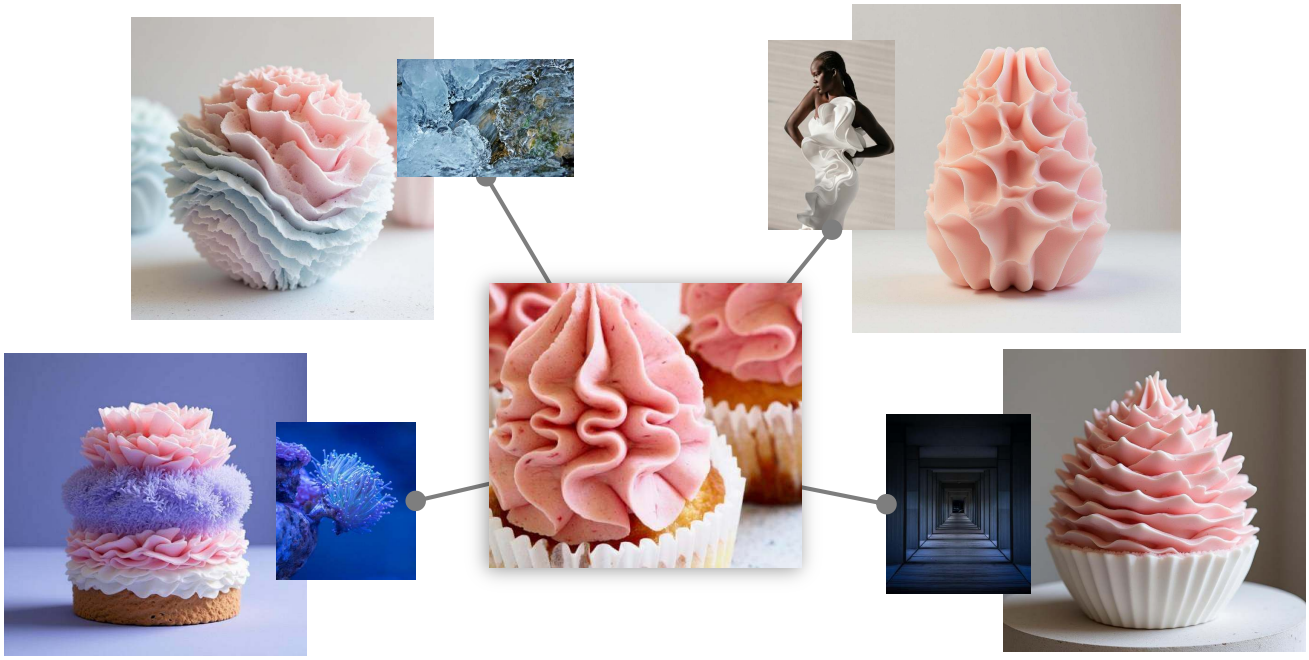

Fig. 1. Our method supports *visual exploration* by generating non-trivial combinations from image pairs. These hybrids blend visual cues across domains to support early-stage ideation, without requiring users to specify their intent in text. Here, a cupcake combined with four different visual references yields diverse transformations, from crystalline mineral layering and coral-like textures (left) to sculptural fabric forms and architectural folded surfaces (right).

While generative models have become powerful tools for image synthesis, they are typically optimized for executing carefully crafted textual prompts, offering limited support for the open-ended visual exploration that often precedes idea formation. In contrast, designers frequently draw inspiration from loosely connected visual references, seeking emergent connections that spark new ideas. We propose Inspiration Seeds, a generative framework that shifts image generation from final execution to exploratory ideation. Given two input images, our model produces diverse, visually coherent compositions that reveal latent relationships between inputs, without relying on user-specified text prompts. Our approach is feed-forward, trained on synthetic triplets of decomposed visual aspects derived entirely through visual means: we use CLIP Sparse Autoencoders to extract editing directions in CLIP latent space and isolate concept pairs. By removing the reliance on language and enabling fast, intuitive recombination, our method supports visual ideation at the early and ambiguous stages of creative work.

1 Introduction

Ideas rarely arrive fully formed. Exploration and inspiration are key to the design process: creators explore by sketching, assembling inspiration boards, and observing artworks, natural phenomena, and abstract forms [Eckert and Stacey 2000; Goldschmidt 1991]. Often, when examining a curated set of references, designers notice unexpected connections between familiar elements. An example of such a connection is shown in Figure 2, where fashion designer Iris van Herpen combines visual aspects of deep-sea organisms and neural structures in non-trivial ways, inspiring her collection of dresses [Iris van Herpen 2020]. These moments of recognition and inspiration lead to new ideas. However, perceiving such hidden visual qualities is challenging and often requires design experience and a creative eye to see beyond obvious connections [Eckert and Stacey 2000; Gentner 1983; Goldschmidt 1991; Tversky 2011].

Recent generative models offer new opportunities to support visual creation [Epstein and Hertzmann 2023; Mazzone and Elgammal 2019], but they are typically used in a very specific way. Most text-to-image models [Google DeepMind 2025; Labs et al. 2025] are designed to execute well-specified ideas through detailed prompts. As a result, they often come into play only after an idea has already been formed and verbalized. This leaves little support for the earlier, exploratory phase of creation, where ideas are still vague, intuitive, and primarily visual [Arnheim 1969; Jonson 2005; Kim and Wilemon 2002].

In this paper, we propose a new perspective on the role of generative models in visual creation: using them as tools for *visual exploration* rather than for producing final, polished images. From this perspective, the output of the model is not an endpoint, but an intermediate representation that can spark new ideas. To support this goal, we introduce *Inspiration Seeds*, a model that takes two

Fig. 2. Dresses from Iris van Herpen's Sensory Seas collection (2020), inspired by a resemblance between deep-sea hydrozoans and neural structures. Surfacing such unique connections is key to producing original designs.

images as input and produces multiple visual combinations designed to surface visual relationships that are difficult to articulate verbally — revealing deep and sometimes surprising connections between the visual qualities of the inputs.

Current image generators and editing tools, even leading ones like Nano Banana [Google DeepMind 2025], tend to produce trivial combinations even when prompted to be "creative", defaulting to straightforward edits as shown in Figure 3 (replacing an earring with a leaf). This is expected given the distribution of typical image edits these models were trained on. Generating more unexpected results typically requires careful prompt engineering and repeated intervention, which runs counter to the fluid, non-verbal nature of visual exploration [Suwa and Tversky 1997]. Our method is designed explicitly to surface non-trivial connections without relying on text: in Figure 3, the leaf's decay pattern, green tones, and aged quality carry over to the subject in unexpected ways. Such outputs can suggest new creative directions, particularly when users do not yet know what they want to create.

A key challenge in learning non-trivial visual combinations is obtaining suitable training data: triplets of two visual concepts and a corresponding non-obvious combination. Manually curating such data at scale is impractical, as it would require identifying and annotating pairs of visual concepts together with non-obvious combinations that are difficult to articulate explicitly. A natural alternative is to generate training data automatically using existing image decomposition methods and train a model to invert this process. However, most existing decomposition methods focus on specific, well-defined relationships, such as decomposing images into explicit

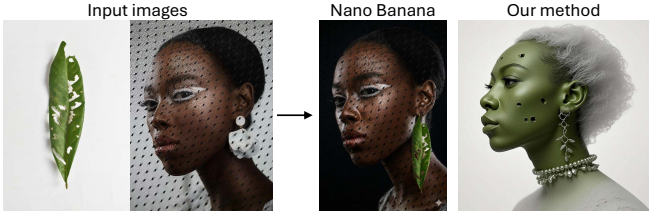

Fig. 3. Trivial vs. non-trivial visual combinations. Given a leaf and a portrait, Nano Banana produces a trivial combination by replacing the earring with a leaf. Our method surfaces deeper connections: the leaf's decay pattern appears in the skin, and its aged quality carries over to the subject.

object-level components [Avrahami et al. 2023] or style–content separation [Frenkel et al. 2024]. While effective for their respective goals, such formulations are inherently limited to a fixed vocabulary of relationships, making them ill-suited for learning the open-ended, non-literal combinations we target.

To go beyond these limitations, we require a data generation pipeline that avoids explicitly specifying the relationship during decomposition, allowing relevant visual aspects to emerge *implicitly* from the image itself. A well-suited conceptual direction is the implicit decomposition proposed in InspirationTree [Vinker et al. 2023b], where the division into concepts is determined during optimization rather than prescribed in advance. However, this approach is designed for single-object decomposition and relies on textual inversion [Gal et al. 2023], requiring multiple images per concept and costly per-image optimization, and often exhibiting optimization-related instabilities.

To address this, we retain the core idea of implicit decomposition while removing the reliance on costly, instable optimization. Our key insight is that the latent representations of pretrained vision–language models already encode multiple, partially disentangled visual concepts within a single image. Specifically, we propose a decompositon approach that uses CLIP Sparse Autoencoders [Daujotas 2024] to extract salient visual factors from each image. These factors are then grouped into coherent visual aspects and then used to define opposing directions in CLIP space that emphasize different visual aspects of the image, enabling a single image to be decomposed into complementary visual views. This decomposition is entirely visual, requires no textual annotations, and provides a fast and scalable foundation for learning non-literal image composition.

We use our decomposition approach to construct a large-scale dataset of non-literal image decompositions and fine-tune an image-conditioned generative model [Labs et al. 2025] to learn the inverse mapping. Leveraging the model's strong visual prior, our approach enables it to capture non-obvious relationships directly from data and generalize to unseen inputs. Importantly, sampling with different random seeds yields varied combinations, each surfacing distinct visual connections. We evaluate on diverse image pairs and show that our method generates non-trivial, visually coherent combinations that reveal deeper relationships between inputs, outperforming leading models that favor literal composition. We additionally propose a description-complexity metric to evaluate this challenging task. We hope this work invites further research into generative models as tools for visual exploration and ideation.

2 Related Work

Design and Modeling Inspiration. The ability to perceive connections between previously unrelated ideas and recombine prior knowledge in new ways is often integral to the design process and to generating new ideas [Bonnardel and Cauzinille-Marmèche 2005; Runco and Jaeger 2012; Wilkenfeld and Ward 2001]. In practice, this process is typically exploratory: designers and artists work with collections of visual elements and references to probe relationships and directions before a concrete concept is fully articulated [Eckert and Stacey 2000]. This phase of ideation is inherently visual and associative, relying on perceived form rather than precise semantic

descriptions. Motivated by this, prior work has proposed computational tools to support ideation by facilitating the organization and comparison of visual material [Ivanov et al. 2022; Kang et al. 2021; Koch et al. 2019, 2020]. While effective for navigating existing examples, such systems primarily operate on curated content.

Image Generation and Personalization. Recent progress in image generation has led to models capable of synthesizing high-quality images that closely follow user instructions [Black Forest Labs 2024; Nichol et al. 2021; Ramesh et al. 2022; Rombach et al. 2022; Saharia et al. 2022; Wu et al. 2025]. These advances have established text-to-image generation as a powerful interface for visual content creation, enabling detailed control over content, style, and composition. However, most existing generative models are optimized for execution rather than exploration. They assume that the desired outcome can be articulated through a well-specified textual prompt, offering limited support for the earlier creative phase in which ideas are still forming and primarily visual [Jonson 2005; Kim and Wilemon 2002]. In such settings, users often lack the language to precisely describe what they seek and instead rely on visual cues, references, and associations. Personalization techniques [Gal et al. 2023; Ruiz et al. 2022] extend text-to-image models by allowing them to incorporate specific user-provided concepts into the generation process. While highly effective for reproducing known concepts, these approaches are not designed to encourage novel visual recombination and are less suited for exploring alternative interpretations or combining multiple visual elements in open-ended ways.

Concept Decomposition. Decomposing images into meaningful visual components is inherently ill-posed, as high-level visual aspects are often entangled and do not correspond to explicit spatial regions or predefined categories. Prior work has explored decomposition along fixed axes — extracting objects via mask-guided personalization [Avrahami et al. 2023; Garibi et al. 2025; Kumari et al. 2023], modeling predefined attributes [Lee et al. 2023; Xu et al. 2024], or separating style from content [Alaluf et al. 2024; Frenkel et al. 2024; Gatys et al. 2015; Ngweta et al. 2023; Shah et al. 2024]. While effective for their intended purposes, these methods rely on predetermined decomposition dimensions rather than discovering new visual factors. InspirationTree [Vinker et al. 2023a] takes a different approach by decomposing a visual concept into unexpected, hierarchical visual attributes. However, it relies on textual inversion [Gal et al. 2023], requiring multiple images of the target concept across different views and backgrounds, as well as hours of optimization per concept. This process is often unstable, making it ill-suited for our setting, where we aim to decompose single images, which may not depict concrete, isolated objects, and to operate efficiently at scale.

Recent advances in mechanistic interpretability offer a promising alternative. Sparse Autoencoders (SAEs), originally proposed to identify monosemantic features in language models [Cunningham et al. 2023], have been applied to CLIP [Radford et al. 2021], decomposing its representations into sparse, interpretable visual factors [Daujotas 2024; Fry 2024; Zaigrajew et al. 2025]. Building on this, our approach leverages SAE-derived features to decompose arbitrary images into interpretable concepts in a single forward pass, without per-image optimization or concept-specific training.

Visually Inspired Generation. Early work on interactive evolutionary computation showed that rich visual artifacts can emerge through iterative selection, without requiring users to explicitly specify their goals [Sims 1991; Takagi 2001]. Picbreeder [Secretan et al. 2008] extended this paradigm to collaborative online settings, enabling open-ended exploration of large design spaces. Similarly, DeepDream [Mordvintsev et al. 2015] revealed that the internal representations of neural networks can serve as a substrate for visual discovery. More recently, generative models have been explored as tools for visual inspiration, helping users discover new ideas through alternative interpretations or novel combinations of visual concepts [Elhoseiny and Elfeki 2019; Hertzmann 2018; Oppenlaender 2022; White 2020]. Several approaches use vision–language guidance to learn novel concepts within broader visual categories [Lee et al. 2024; Richardson et al. 2024], while others focus on visually conditioned generation, where models are guided by image embeddings rather than text. Methods such as IP-Adapter [Ye et al. 2023] enable manipulation in embedding space and have been used to define composition rules over visual concepts [Dorfman et al. 2025; Richardson et al. 2025a,b]. However, these approaches typically rely on predefined operations, leaving open the challenge of enabling open-ended, non-literal visual exploration driven purely by visual input.

3 Preliminaries

Sparse Autoencoders for CLIP. Neural networks often exhibit *polysemanticity*, where individual neurons respond to multiple, semantically distinct concepts. This phenomenon arises from *superposition*, in which more features are encoded than there are available representational dimensions [Cunningham et al. 2023]. As a result, individual activation dimensions are difficult to interpret in isolation. Sparse Autoencoders (SAEs) address this by learning an overcomplete and sparse factorization of activations into interpretable features. Given an activation vector $\mathbf{a} \in \mathbb{R}^n$ from a network layer, an SAE learns an encoder $\mathbf{W}_{\text{enc}} \in \mathbb{R}^{m \times n}$ and a decoder $\mathbf{W}_{\text{dec}} \in \mathbb{R}^{n \times m}$ where $m \gg n$:

$$\mathbf{h} = \sigma(\mathbf{W}_{\text{enc}}\mathbf{a} + \mathbf{b}_{\text{enc}}), \quad \hat{\mathbf{a}} = \mathbf{W}_{\text{dec}}\mathbf{h} + \mathbf{b}_{\text{dec}}, \tag{1}$$

The SAE is trained with a sparse reconstruction loss $\mathcal{L}_{\text{SAE}} = \|\mathbf{a} - \hat{\mathbf{a}}\|_2^2 + \lambda\|\mathbf{h}\|_1$, which encourages $\mathbf{h}$ to activate only a small subset of features for any given input. Each column of $\mathbf{W}_{\text{dec}}$ corresponds to a learned feature direction, while the sparse coefficients $\mathbf{h}$ indicate which features are present in the activation. When applied to CLIP embeddings, SAEs reveal monosemantic visual concepts such as textures, shapes, and styles, which can be used to decompose images, visualize features through diffusion rendering, and steer generative models [Fry 2024; Zaigrajew et al. 2025].

FLUX.1 Kontext. FLUX.1 Kontext [Labs et al. 2025] is a rectified flow model that unifies image generation and editing. Kontext processes concatenated image and text token sequences through a Multimodal Diffusion Transformer [Esser et al. 2024; Peebles and Xie 2023], supporting both generation and in-context editing within a single architecture. Its strong prior on both generation and context understanding makes it a natural candidate for our base model.

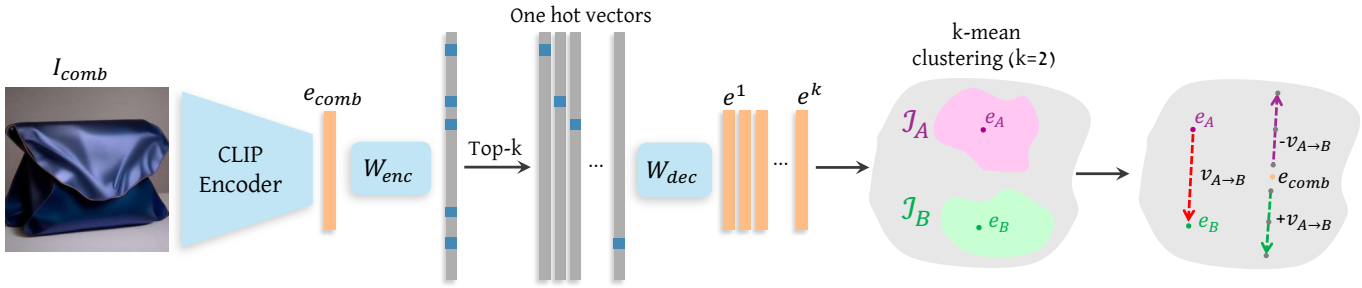

Fig. 4. Overview of our image decomposition pipeline. Given an image I_{comb}, we encode it via CLIP and pass the embedding through an SAE encoder W_{enc}. We retain the top-k activations as one-hot vectors, and decode them back to CLIP space via W_{dec}. We then cluster the resulting vectors into two groups using k-means. The editing direction $v_{A\to B}$ is computed as the difference between cluster centroids. Moving e_{comb} in opposite directions along this axis and decoding via Kandinsky yields two images I_A and I_B that emphasize distinct visual aspects of the original image.

4 Method

Our goal is to design a model that takes two images as input and generates multiple visual combinations that reveal non-trivial connections between them, without relying on textual supervision or user-provided instructions. We formulate this task as training an image-to-image model $f_\theta(I_A, I_B) \to I_{\text{comb}}$, which receives two images and outputs a combined image. We fine-tune Flux.1 Kontext [Labs et al. 2025], a large pretrained model for image generation and editing, to perform this visual composition task.

A central challenge is obtaining suitable training data: triplets $(I_A, I_B, I_{\text{comb}})$, where the combination reflects a meaningful visual relationship rather than a superficial one. Manually constructing such data at scale is impractical. Our key insight is to invert this problem: instead of searching for image pairs that combine well, we start from visually rich images and decompose them into two constituent visual aspects. The original image then serves as a ground-truth combination, providing natural supervision for training.

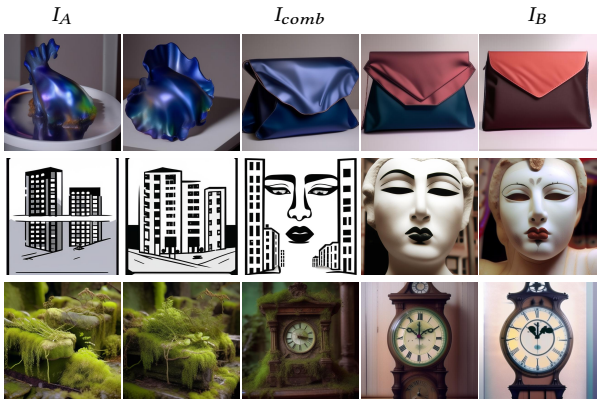

Fig. 5. Examples of decomposed triplets. Each row shows two variations (I_A, I_B, I_{comb}) derived from a source image I_{comb} using our CLIP SAE decomposition. The decomposition separates distinct visual aspects.

4.1 Image Pool Construction

Our decomposition approach requires images that intentionally combine multiple distinct visual aspects within a single image. Typical single-object photographs may vary in color, pose, or shape, but they rarely contain several independently meaningful visual ideas that can later be separated and recombined. To obtain such images at scale, we generate a diverse image pool with two complementary strategies. First, we use templated prompts that explicitly specify multiple visual properties (e.g., material, color, shape, and context) to produce "multi-attribute" images. Second, we start from intentionally vague prompts (e.g., "a place that never was") and use large language models (LLMs) to expand them into richer descriptions before generation. We expand each vague prompt into multiple distinct visual interpretations, yielding semantically related but visually diverse images. This results in a pool of visually rich images designed to support decomposition into non-trivial visual aspects. Having constructed images that intentionally bundle multiple visual aspects, we next decompose each image into its constituent aspects.

4.2 Image Decomposition via CLIP SAEs

Decomposing a single image into multiple meaningful visual aspects is a highly non-trivial task, as there is no canonical way to separate an image into constituent components and high-level visual aspects are often entangled. While this problem has been explored in prior work [Avrahami et al. 2023; Frenkel et al. 2024; Kumari et al. 2023; Vinker et al. 2023a], most existing approaches rely on text-to-image personalization optimization [Gal et al. 2023; Ruiz et al. 2022], which is time-consuming and typically decomposes images into explicit sub-objects rather than more abstract or non-obvious visual aspects.

To address this, we formulate decomposition as controlled editing in CLIP latent space, where linear directions correspond to meaningful visual transformations [Radford et al. 2021]. Rather than relying on predefined attributes or textual supervision, our approach derives image-specific decomposition axes directly from the visual content of each image.

Given a source image I_{comb} from the set described above, our goal is to produce two images I_A and I_B that each emphasize a distinct visual aspect of I_{comb}, such that I_{comb} can be interpreted as a combination of the two. Examples of such decompositions are illustrated in Figure 5. Our pipeline is illustrated in Figure 4. Let $e_{comb} = \text{CLIP}(I_{comb})$ denote the CLIP embedding of the input image. Our goal is to find an editing direction $v_{A\to B}$ that separates two dominant visual aspects within e_{comb}, and obtain two edited embeddings by moving e_{comb} in opposite directions along this axis

(Figure 4, right). These embeddings can then be decoded via a CLIP-to-image generator such as Kandinsky [Razzhigaev et al. 2023] into images I_A and I_B, with I_{comb} lying conceptually between them.

To identify such editing directions in practice, we leverage CLIP Sparse Autoencoders (SAEs) [Daujotas 2024], which expose interpretable visual attributes from CLIP embeddings. Given e_{comb}, we first encode it using the SAE encoder W_{enc} to obtain a sparse activation vector, and retain the top-k activated entries. This yields a set of k one-hot vectors, each corresponding to a dominant potential visual factor present in the image. Because CLIP SAE features are not fully disentangled, highly activated features often capture closely related visual attributes with subtle variations. Therefore, rather than selecting individual features, we group them into higher-level visual aspects. Specifically, we decode each one-hot vector back into CLIP space using the SAE decoder W_{dec}, producing a set of vectors $\{e^1, \ldots, e^k\}$ shown in orange. We cluster these vectors using k-means with k=2, yielding index sets $\mathcal{I}_\mathcal{A}$ and $\mathcal{I}_\mathcal{B}$. To improve coherence, we discard vectors near the cluster boundary.

From the filtered clusters we can now compute an editing direction as the difference between the two cluster centroids:

$$e_A = \frac{1}{|\mathcal{I}_\mathcal{A}|} \sum_{j \in \mathcal{I}_\mathcal{A} e^j} e^j, \quad e_B = \frac{1}{|\mathcal{I}_\mathcal{B}|} \sum_{j \in \mathcal{I}_\mathcal{B} e^j} e^j, \quad v_{A \to B} = e_B - e_A. \quad (2)$$

We then produce two edited embeddings by moving the original embedding in opposite directions:

$$e_{comb \to A} = e_{comb} - \lambda\, v_{A \to B}, \quad e_{comb \to B} = e_{comb} + \lambda\, v_{A \to B}. \quad (3)$$

Finally, we generate images I_A and I_B from these embeddings using the Kandinsky model [Razzhigaev et al. 2023], which was designed to support CLIP embedding conditioning. The resulting triplet (I_A, I_B, I_{comb}) provides natural, optimization-free supervision for training our combination model.

4.3 Training

Our final synthetic image pool consists of 2085 images from which we produce 2085 triplets (I_A, I_B, I_{comb}) using our decomposition pipeline. Using this set we can now train a model to perform the inverse task: given two images, produce a combination that captures visual aspects of both. In practice, we fine-tune Flux.1 Kontext [Labs et al. 2025] using LoRA with a rank of 32. The input images are resized to 512×512 pixels and placed on a 1024×1024 canvas: I_A in the top-left corner and I_B in the bottom-right, with the remaining area filled with white. The model is trained to generate I_{comb} conditioned on this canvas. To avoid textual bias during training and inference, we use a fixed prompt: *"Combine the element in the top left with the element in the bottom right to create a single object inspired by both of them."* We tune the model for 15k steps using the Ostris AI-Toolkit [2025]. At inference, given any two images, the model can generate multiple combinations by varying the random seed, surfacing different visual relationships between the inputs.

5 Experiments

5.1 Visual Exploration Results

We first present qualitative examples that illustrate how our method can facilitate visual exploration. Figures 1 and 6 to 8, show results generated by our method. In Figure 1, a cupcake combined with four different references (a mineral texture, a flowing dress, a sea anemone, and a receding corridor) yields distinct visual transformations, with the frosting adopting crystalline layering, fabric-like folds, organic branching, or architectural geometry depending on the input. In Figure 6 we show how our method can implicitly produce different interpretations of the given input images when varying the seed, a property well-suited to exploratory workflows.

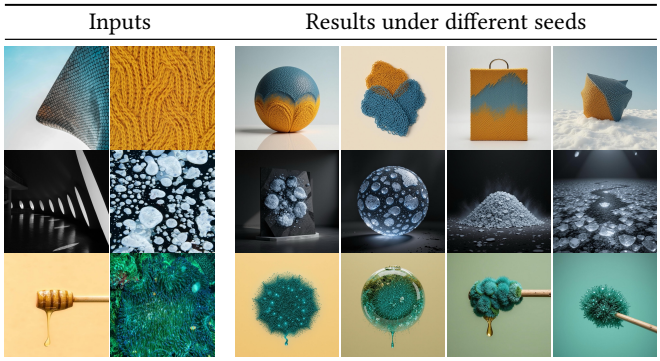

Fig. 6. Visual Combinations under different seeds. For the same pair of input images our model can produce different visual combinations just by varying the seed, without any explicit guidance.

Flux.1 Kontext and Qwen-Image are open source models. Nano Banana, Reve, and ChatGPT are closed source commercial models. Baseline methods often produce trivial combinations: direct copying of the inputs (e.g., Flux reproducing the input layout in the first two rows and copying the grid input in the third row), or object insertion (e.g., Nano Banana inserting the insect intro the mushrooms scene in the first row). We note that ChatGPT often produces better combinations though its style often tend to be over polished or drametic

In Figure 8 and Figure 7, we present additional results in the form of an exploration canvas, reflecting how we envision the method being used in practice. This canvas illustrates a potential workflow where a user might collect reference images, combine them in different pairings, and branch out from promising results. Figure 8 shows diverse input images (a honey dipper, a woven mesh, a mineral texture, a flock of birds, a strawberry mushroom, a shell) connected to grids of multiple outputs generated from their pairings. For example, a honey dipper paired with underwater flora transforms into mossy organic forms (top left); paired with a woven mesh, it takes on golden shell-like qualities (middle left). Using different seeds provides diversity, which is key to supporting exploration: rather than producing a single "correct" combination, the model generates a space of options that users can browse, allowing unexpected connections to emerge without requiring users to articulate what they are looking for. More results and an interactive demo are available in the supplementary material.

5.2 Evaluating Visual Combinations

Here we evaluate our method's ability to produce meaningful, non-trivial visual combinations. We curate a benchmark of 41 images spanning six categories (architecture, fashion, food, nature, sea creatures, and other), sourced from Pexels, to cover a range of concepts,

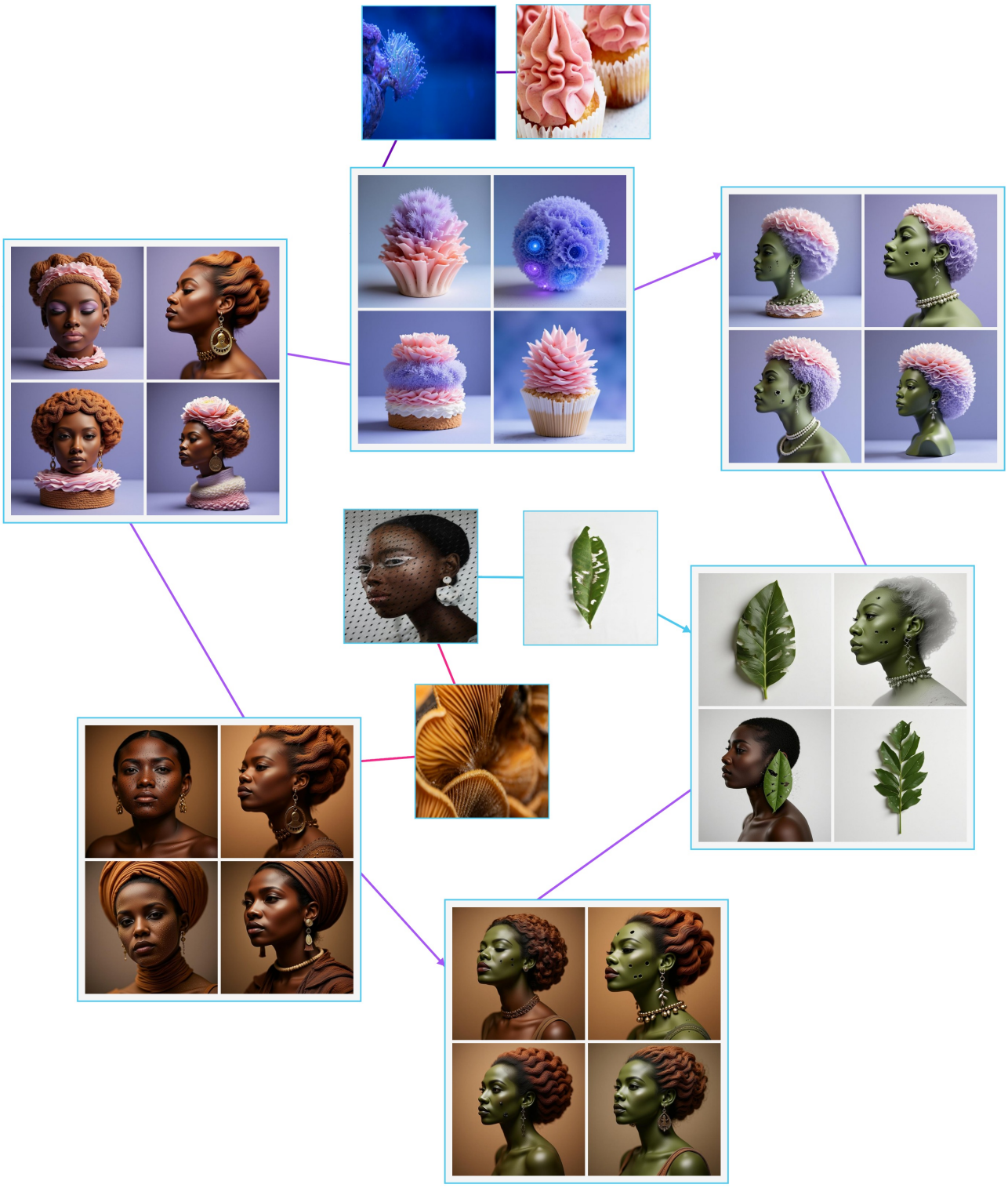

Fig. 7. Iterative exploration. Outputs can serve as inputs for further combination. A cupcake paired with coral produces frosting with organic, anemone-like texture (top grid). A portrait paired with a leaf yields green skin and botanical patterns (middle grid); paired with jellyfish produces bioluminescent figures (bottom left grid); paired with fungi creates warm tones and sculptural, ruffled hair (bottom middle grid). The rightmost grids show further iterations: combining the portrait-leaf output with the cupcake-coral output produces figures with green skin and fluffy pink hair (top right); combining with the fungi output yields warm-toned portraits with layered, textured hair (bottom right). Each iteration accumulates visual qualities from multiple sources.

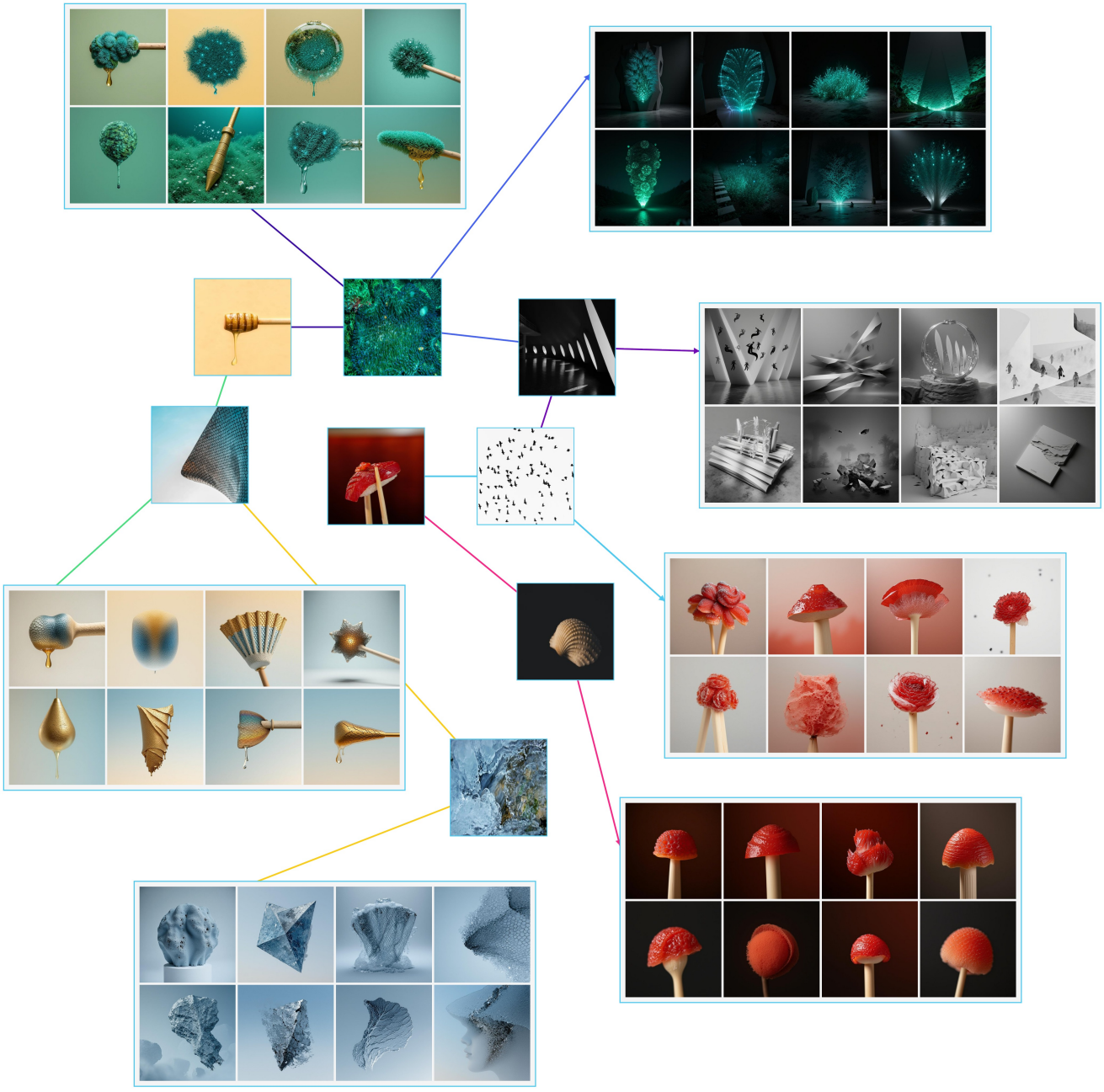

Fig. 8. Exploration canvas. We present results in the format of an infinite canvas, reflecting how we envision the method being used in practice. Input images (center) are paired with different references, producing grids of outputs generated with varying random seeds. The canvas structure allows users to browse combinations, compare variations, and branch out from promising results, supporting open-ended exploration rather than converging on a single output.

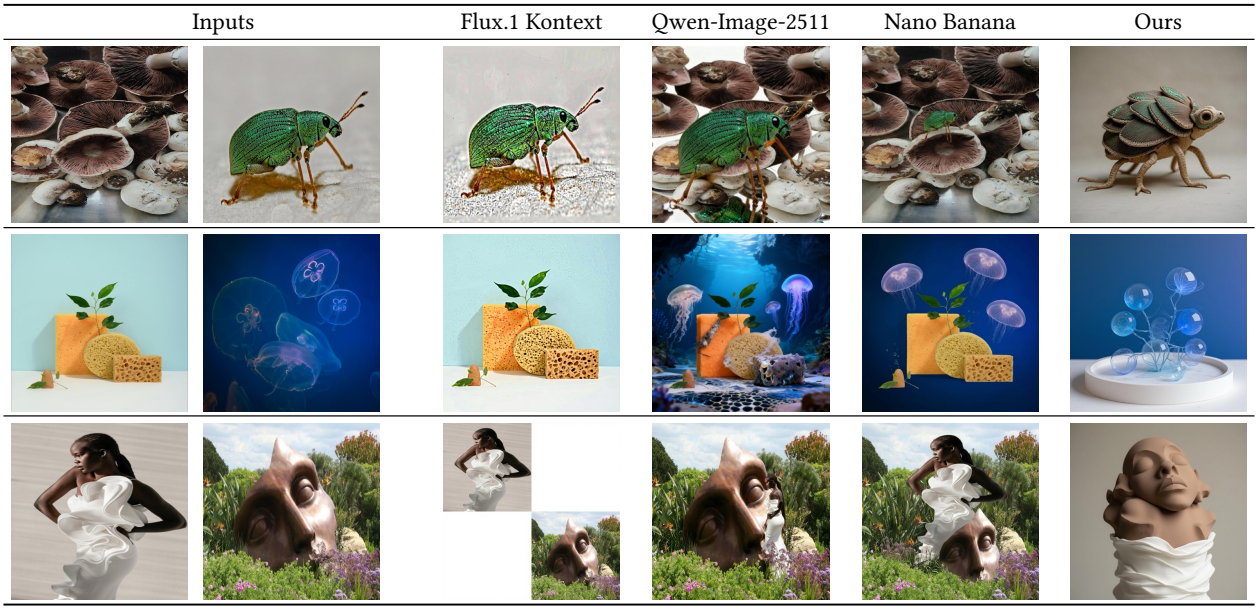

Fig. 9. Qualitative comparison of visual combinations. Baseline methods often produce trivial combinations: direct copying of the inputs (e.g., Flux reproducing the input layout in the first two rows and copying the grid input in the third row), or object insertion (e.g., Nano Banana inserting the insect intro the mushrooms scene in the first row). In contrast, our method produces images in which visual cues from both inputs are integrated into a single coherent form.

styles, structures, and materials. We randomly sample 99 cross-category pairs, ensuring each image appears in at least one pair.

Since no existing method is explicitly designed to generate non-trivial visual combinations from image pairs, we compare against the strongest publicly available image-conditioned generation models. Specifically, we evaluate Flux.1 Kontext [Labs et al. 2025], a large-scale image editing model that also serves as our backbone; Qwen-Image-2511 [Wu et al. 2025], a recent multimodal model with strong visual understanding capabilities; and Nano Banana [Google DeepMind 2025], Google's image generation and editing model. For Flux.1 Kontext, we use the same constant input prompt used in our training, matching its single input design. For Qwen and Nano Banana, we provide both images along with the prompt: *"Combine the two images into a novel and non-trivial image inspired by them."* For all methods, we generate four random outputs per pair using different seeds, resulting in 396 images per method in total.

Representative results are shown in Figure 9. For visualization clarity, we display one output per method for each input pair. All generated samples, including all seeds, are provided in the supplementary material. Flux.1 Kontext tends to copy the input images, either completely or by reproducing its input grid-like arrangement. Qwen-Image-2511 often defaults to trivial combinations. Nano Banana performs best among the baselines, however, it often defaults to object-level placement without transferring deeper visual qualities. In contrast, our method produces coherent, non-trivial combinations, integrating visual aspects from both inputs. The beetle takes on the mushroom's layered patterns, the sponge and jellyfish merge into delicate, bubble-like forms, and the portrait and sculpture blend into a figure where skin and fabric share the same materiality. These connections are not immediately obvious, they require a close look, and invite interpretation, which is what makes them useful for creative work. They surface relationships a user might not have thought to look for.

Quantitative Evaluation. Standard perceptual similarity metrics such as CLIP cosine similarity [Radford et al. 2021] or DreamSim [Fu et al. 2023] reward visual similarity. In our setting, this means outputs that simply preserve or insert elements from the inputs score higher than those that transform and recombine them. Moreover, such metrics are not designed to measure whether a combination is non-trivial or unique. We therefore propose using *description complexity* as an alternative measure. We observe that trivial combinations can often be explained in a few words ("place object A into scene B"), whereas non-trivial combinations require longer descriptions to articulate. This observation aligns with research linking description length to complexity. Specifically, Kolmogorov complexity formalizes the idea that an object's complexity corresponds to the length of its shortest description [Kolmogorov 1965; Li and Vitányi 1997], and Sun and Firestone [Sun and Firestone 2022] showed that verbal description length tracks the information-theoretic complexity of visual stimuli.

To apply this, we prompt Gemini 2.5 Flash to describe how each output image could be reconstructed from its two source images, using a fixed instruction format across all methods (see supplementary material for more details). We use word count as a proxy for the complexity of the visual relationship. The average word counts across all 99 image pairs are shown in Table 1. Our method elicits longer descriptions on average compared to all baselines. Figure 10 illustrates the textual descriptions obtained by the LLM across different methods.

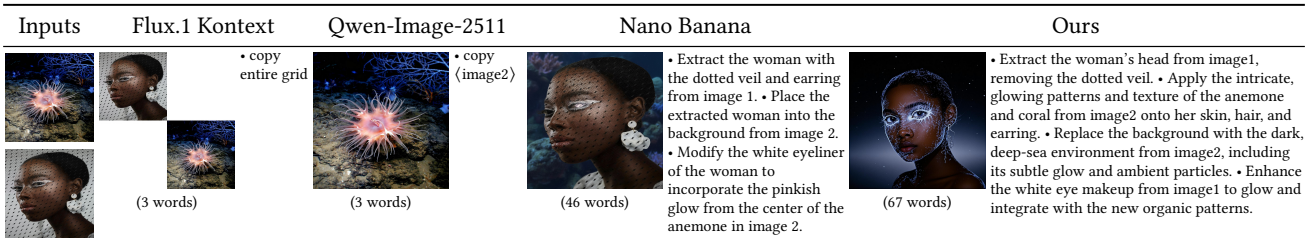

Fig. 10. Description complexity analysis. We show the LLM's descriptions for reconstructing the target image from the two inputs. One can see that as the connection becomes more complex and non-literal, the LLM naturally scales to longer and more sophisticated descriptions.

Table 1. Caption length comparison for combination complexity. We measure the word count of VLM-generated descriptions explaining how to recreate the output from the inputs. Higher word counts indicate more complex, non-trivial combinations. We also report the percentage of outputs classified as trivial patterns.

Method	Word Count	Copy	Insertion	Split
Flux.1 Kontext	23.5 ± 21.4	2.8%	0.3%	85.4%
Qwen-Image	37.4 ± 19.2	16.2%	18.9%	10.6%
Nano Banana	42.9 ± 15.6	9.1%	19.7%	0.3%
Ours	**54.8 ± 12.5**	**2.3%**	**0.0%**	**1.5%**

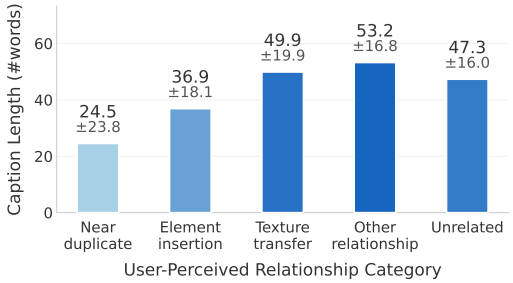

Fig. 11. User study results. Simple relationships such as duplication or insertion require less words than more complex ones such as texture transferring or other non-canonical relationships.

We additionally analyze the types of relationships recognized by Gemini, counting observable patterns such as copying (output nearly identical to one input), insertion (placing one element into the other scene), or split composition (inputs placed side by side or in a grid), as demonstrated in Figure 9. As shown in Table 1, Flux.1 Kontext defaults to split compositions in most cases, while Qwen and Nano Banana often resort to insertion. Our method rarely triggers any of these categories, indicating that the combinations it produces do not reduce to simple operations.

User Study. To provide additional support that description length serves as a meaningful proxy for combination complexity, we conduct a user study with 35 participants. Each participant was shown an output image alongside its two inputs and asked to classify the relationship between them. The options were: (1) near-duplicate; (2) element insertion; (3) texture or structure transfer; (4) other relationship not captured by the above; and (5) unrelated. We sampled 25 outputs stratified by description length, comprising 11 images from our method and 7 each from Nano Banana and Qwen.

In Figure 11 we report the average description length of images assigned by participants to each category. If description length captures combination complexity, we would expect images classified as more complex relationships to have longer descriptions. Indeed, description length increases with combination complexity. Images associated with trivial relationships such as duplication or insertion are more easily described, whereas texture and structure transfer requires explicit specification of which visual properties are extracted and how they are mapped onto another structure. Outputs categorized as "other relationship" similarly demand longer explanations, as they involve transformations that are not captured by predefined operations.

5.3 Decomposition Results

Our decomposition technique is central to our approach as it determines what kinds of relationships the model learns to combine. InspirationTree [Vinker et al. 2023b] is the only existing method for implicit decomposition beyond style-content separation. However, it is designed for decomposing images of single objects and relies on textual inversion, requiring 4-5 images of the same concept from different viewpoints, over an hour of optimization per concept, and multiple runs to handle instability (as noted by the authors). This makes it impractical for our setting, where we aim to decompose arbitrary images efficiently. We therefore construct an alternative, feed-forward baseline to evaluate our approach. Given an input image, we prompt Qwen3-VL-8B-Instruct to describe two possible inspiration sources that could have been combined to form it. Then, we generate images from these descriptions using Flux.1 Kontext in two settings: from text alone (T2I), and with the input image as conditioning (I2I). We illustrate the results of these two baselines and InspirationTree in Figure 12. Our method significancy improves over the baselines while producing meaningful decomposition results similar to those of InspirationTree from just a single image and in a feed-forward manner.

Next, we evaluate our method compared to the proposed baselines on 915 images from our synthetic image pool (described in Section 4.1). InspirationTree is omitted from this evaluation due to its significant runtime. Intuitively, A good decomposition should produce two components that are both related to the input but distinct from each other. Thus, we compute DreamSim [Fu et al. 2023] similarity between each component and the input, and report their

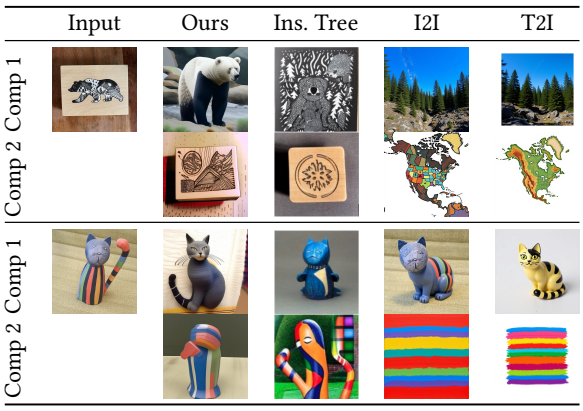

Fig. 12. Decomposition results comparison. Given an input image, we decompose it into two components using four methods. Our approach produces components that capture distinct visual aspects while maintaining semantic relevance. Inspiration Tree relies on time-consuming optimization and requires multiple input images of the concept to converge. T2I and I2I methods often fail to adhere to the visual qualities of the input.

Table 2. Decomposition quality measured by DreamSim similarity. We report similarity between each component and the input (Comp1/Comp2 ↔ Orig), similarity between components (Comp1 ↔ Comp2), and the harmonic mean of component-to-input similarities. A good decomposition should have high harmonic mean (both components related to input) and low component similarity (components are distinct).

Method	Comp1 ↔ Input	Comp2 ↔ Input	Comp1 ↔ Comp2	Harmonic Mean ↑
Base T2I	0.51 ± 0.18	0.25 ± 0.11	0.24 ± 0.10	0.31 ± 0.10
Base I2I	0.69 ± 0.24	0.32 ± 0.17	0.28 ± 0.15	0.40 ± 0.16
Ours	**0.55 ± 0.14**	**0.56 ± 0.14**	**0.31 ± 0.14**	**0.53 ± 0.10**

harmonic mean, which penalizes decompositions where one component matches but the other does not. The results in Table 2 show that our method achieves the highest harmonic mean, with high similarity of both generated components to the input while achieving low average similarity between the two components themselves. The baselines tend to produce one similar component and one unrelated.

6 Discussion, Limitations, and Conclusions

We presented *Inspiration Seeds*, a method for generating non-trivial visual combinations from pairs of images. Unlike existing approaches that execute well-specified ideas, our method is designed to support the earlier, exploratory phase of visual work — surfacing unexpected connections between visual concepts without requiring users to articulate what they are looking for.

Central to our approach is a decomposition technique using CLIP SAEs, which enable automatic generation of training data without predefined relationship categories. This allows our model to learn open-ended visual combinations rather than being restricted to fixed transformations like style transfer or object insertion. Finally, we introduced a new evaluation framework based on description complexity, grounded in research linking description length to cognitive complexity. Our experiments show that our method produces combinations that require richer descriptions than those generated by other methods, indicating deeper integration of visual aspects.

While our method paves the way toward supporting visual exploration through non-trivial image combinations, it naturally has limitations. First, it currently supports only two input images; extending to multiple inputs could enable richer combinations, better reflecting how designers draw from many references simultaneously. Second, users currently have limited control over the combination; enabling them to specify additional controls such as how much of each input to incorporate through a continuous axis would allow for better interaction. Moreover, our method still takes around 30s per generation. Ideally being able to generate results in mere seconds could significantly benefit to the interactive experience.

We hope this work opens new directions for generative models that support exploratory settings in visual domains, enhancing visual ideation while keeping the human creator at the center.

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Inspiration Seeds: Learning Non-Literal Visual Combinations for Generative Exploration Supplementary Material

		13
A	Visual Exploration Interface	13
B	Implementation Details	13
C	Dataset Construction	14
D	Additional Results	14
D.1	Comparison with Baselines	14
D.2	Decomposition Results	14
E	Comparison with CLIP Space Interpolation	14
F	Description Complexity Evaluation	14
G	User Study Details	15

A Visual Exploration Interface

We provide an interactive demonstration of our exploration canvas, illustrating how our method can support ideation and exploration in visual space. The demo allows users to freely combine images from a curated gallery and observe the resulting visual combinations generated by our model. To use the interactive demo, **extract the supplementary ZIP file, navigate to the `demo` directory, and open `index.html` in a web browser.** Figure 13 illustrates the interaction workflow: (1) click an image in the gallery to select it, (2) the selected image is placed on the canvas, (3) available images that can be combined with the selection are highlighted in green—select one to create a pair, (4) the resulting combinations appear on the canvas, and (5) click on any result to use it as input for further exploration, or select a new image from the gallery to continue. You can drag images and results to organize them in any structure you prefer.

More results of our exploration canvases are provided in Figures 16, 17, 19 and 20.

B Implementation Details

Our model builds upon FLUX.1 Kontext [Labs et al. 2025]. We condition generation on two input images by creating a 1024×1024 white canvas and placing each input image (resized to 512×512) in the top-left and bottom-right corners. To remove textual bias during training and inference, we use a fixed prompt: "combine the element in the top left with the element in the bottom right to create a single object inspired by both of them".

We fine-tune using LoRA [Hu et al. 2022] with rank 32 for linear layers and rank 16 for convolutional layers. We use AdamW with learning rate 10^{-4} and batch size 1 for 15,000 steps on a single NVIDIA L40s GPU (approximately 24 hours). Training is performed using the Ostris AI-Toolkit [Ostris AI-Toolkit Contributors 2025].

Given two input images, we arrange them in the 2×2 grid and generate using the fixed prompt from training. Generation takes approximately 34 seconds per image on a single NVIDIA L40s GPU.

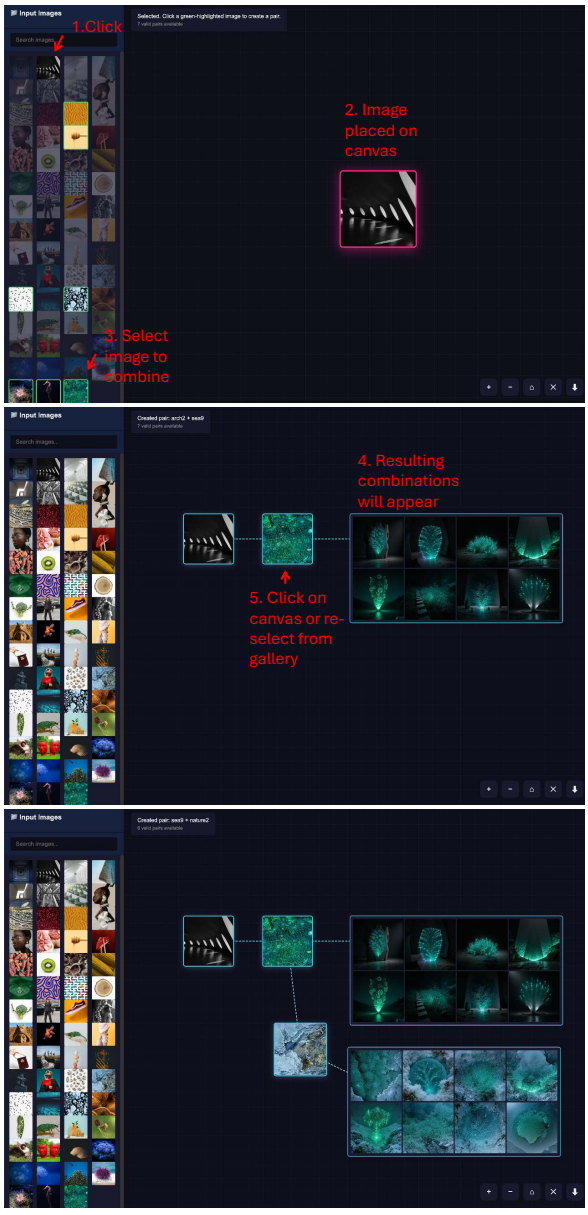


Fig. 13. Interactive demo workflow. Users select an image from the gallery (1), which is placed on the canvas (2). Available pairing options are highlighted in green (3). After selecting a second image, the resulting visual combinations appear on the canvas (4). Users can click any result to continue exploring, or select a new image from the gallery (5).

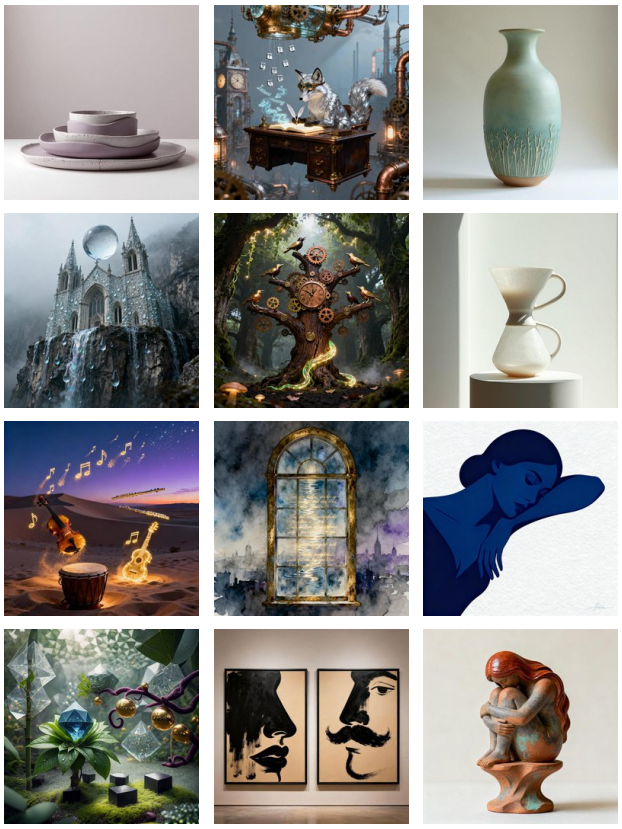


Fig. 14. Examples of images in our data pool. These images are designed to contain multiple distinct visual aspects and will be decomposed by our SAE-based decomposition technique.

C Dataset Construction

To create diverse source images for decomposition, we use several text-to-image generation models: Flux.1 Dev [Black Forest Labs 2024], Fibo [Gutflaish et al. 2025], Reve [Reve AI 2024], and Seedream4 [Seedream et al. 2025] with prompts generated both by using templated prompts, and by using LLMs to expand short, vague prompts to structured prompts, as described in the main paper.

Some examples of our generated data are shown in Figure 14.

D Additional Results

D.1 Comparison with Baselines

In the main paper, we show one output per method for visualization clarity. Here we provide the complete comparison results, showing all four seeds generated per method for each input pair, along with many more examples in Figures 24 to 28.

D.2 Decomposition Results

We provide additional decomposition examples comparing our SAE-based approach to the T2I and I2I baselines described in the main paper in Figure 29.

E Comparison with CLIP Space Interpolation

We compare our composition method against a naive baseline of interpolating in CLIP space. Given two input images, the baseline embeds each into CLIP space, computes the mean of their embeddings, and uses Kandinsky [Razzhigaev et al. 2023] to generate an image from this averaged representation.

As shown in Figure 21, the CLIP interpolation baseline tends to produce blended or averaged results that are not always visually coherent.

F Description Complexity Evaluation

As described in the main paper, we use description complexity as a proxy for measuring the non-triviality of visual combinations. We prompt Gemini 2.5 Flash [Gemini Team, Google 2025] to describe how each output image could be reconstructed from its two source images, then measure the word count of the response. We use the following prompt for evaluating our method, Nano Banana and Qwen-Image-2511:

```
The first two images inspired the third.
Describe briefly how you would recreate the
output using only the two inputs.
Use short bullet points, not paragraphs.
Maximum 5 bullets total, but you do not
have to use them all.
Notes:
* Use "*" to denote bullets. Your answer
should include only bullet points, no free
text.
* Be concise when possible.
* If the output image is very similar to
one of the inputs you can just say "copy
<image1>/<image2>" accordingly.
* Examples of instructions you can use:
"place object from <image1> in the scene
from <image2>", "copy <image1>", "copy <image2>",
"use the object from <image1> and the texture
from <image2>". These are just examples,
you can write your own instructions.
```

For Flux.1 Kontext, we pass the two inputs as a single image with the same grid structure used for inference, and slightly modify the prompt to describe this structure:

```
The first image is 2x2 grid with two images
in the top-left and bottom-right quadrants,
which inspired the second image. Describe
briefly how you would recreate the output
using only the two images in the grid.
Use short bullet points, not paragraphs.
Maximum 5 bullets total, but you do not
have to use them all.
Notes:
* Use "*" to denote bullets. Your answer
should include only bullet points, no free
text.
* Be concise when possible.
* If the output image is very similar to
```

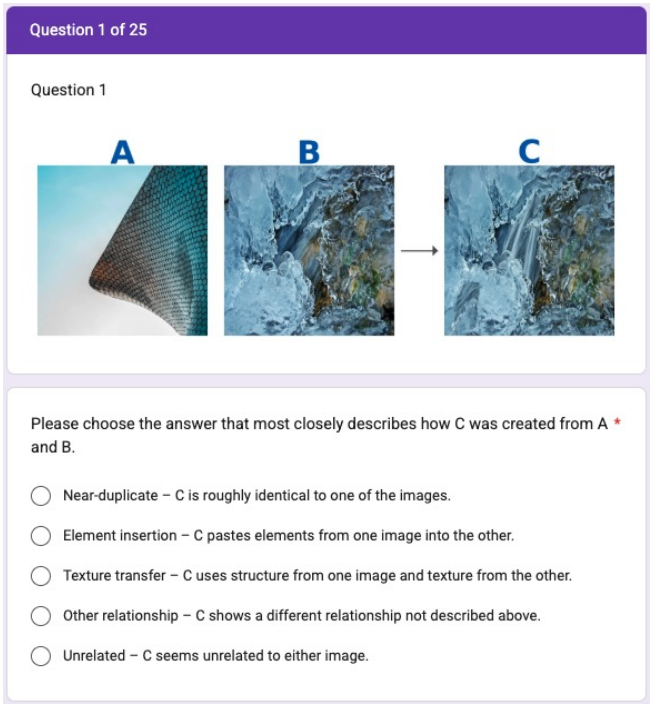


Fig. 15. User study interface. Participants viewed an output image alongside its two inputs and classified the relationship between them.

```
one of the inputs you can just say “copy
<image1>/<image2>” accordingly.
* Examples of instructions you can use:
“place object from <image1> in the scene
from <image2>”, “copy <image1>”, “copy <image2>”,
“copy entire grid”, “use the object from
<image1> and the texture from <image2>”.
These are just examples, you can write your
own instructions.
```

We show examples of the resulting descriptions for the outputs of different methods in Figures 22 and 23.

G User Study Details

We recruited 35 participants through university mailing lists and personal networks. Participants completed the study via Google Forms. For each of 25 trials, participants were shown an output image alongside its two input images and asked to classify the relationship between them. The study took approximately 12 minutes to complete. Figure 15 shows the study interface.

Participants selected from five options describing how the output relates to the inputs:

(1) **Near-duplicate** — the output is roughly identical to one of the input images.
(2) **Element insertion** — elements from one image are pasted into the other.
(3) **Texture transfer** — structure from one image combined with texture from the other.
(4) **Other relationship** — a relationship not captured by the above categories.
(5) **Unrelated** — no apparent connection to either input.

We sampled 25 output images stratified by description length to ensure coverage across the complexity spectrum, comprising 11 images from our method and 7 each from Nano Banana and Qwen-Image-2511. We omit Flux.1 Kontext from the study as it has a different input format which it tends to copy and would require different classification options, and the other methods produced superior results.

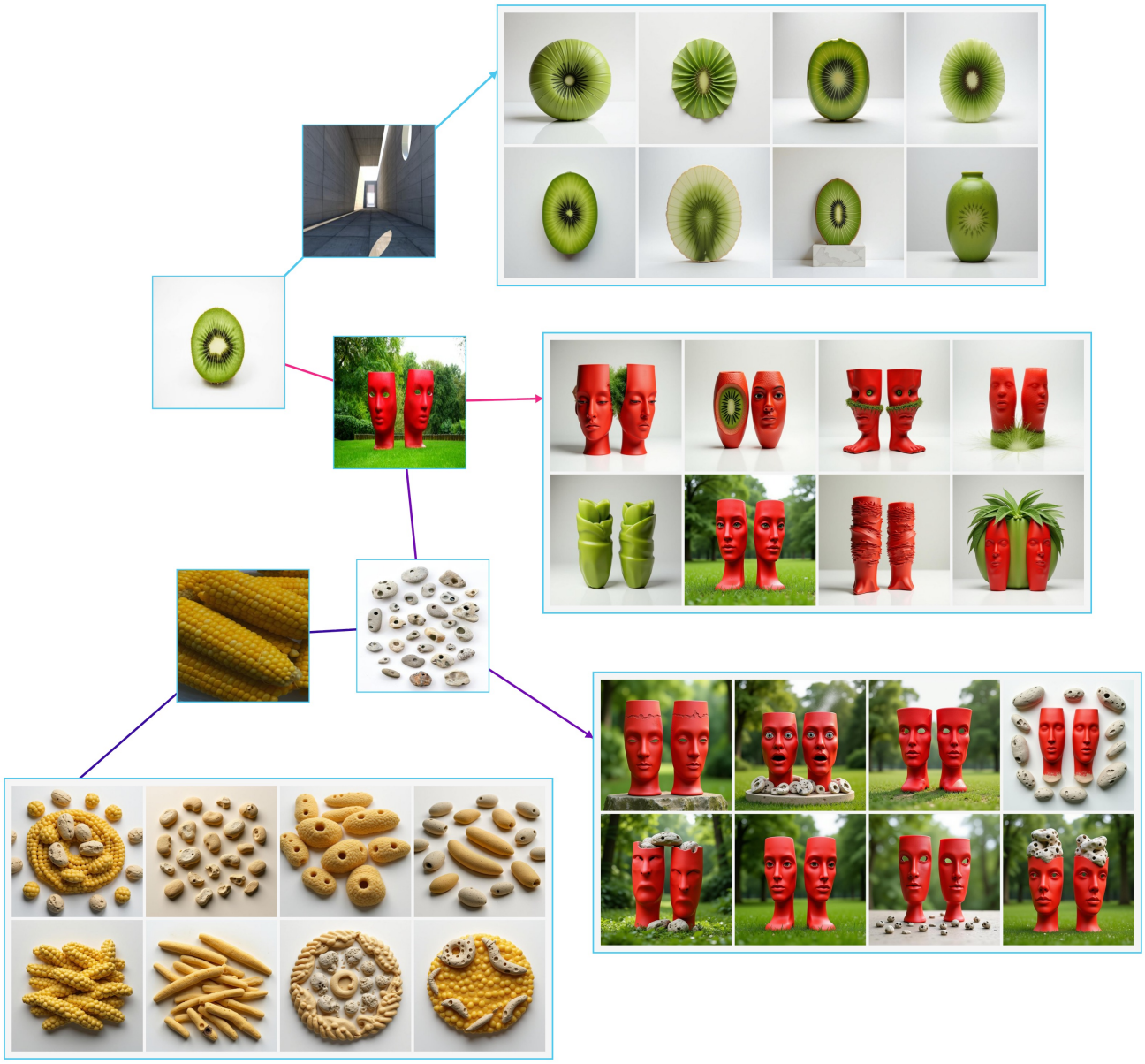


Fig. 16. Exploration canvas showing visual combinations generated by our method.

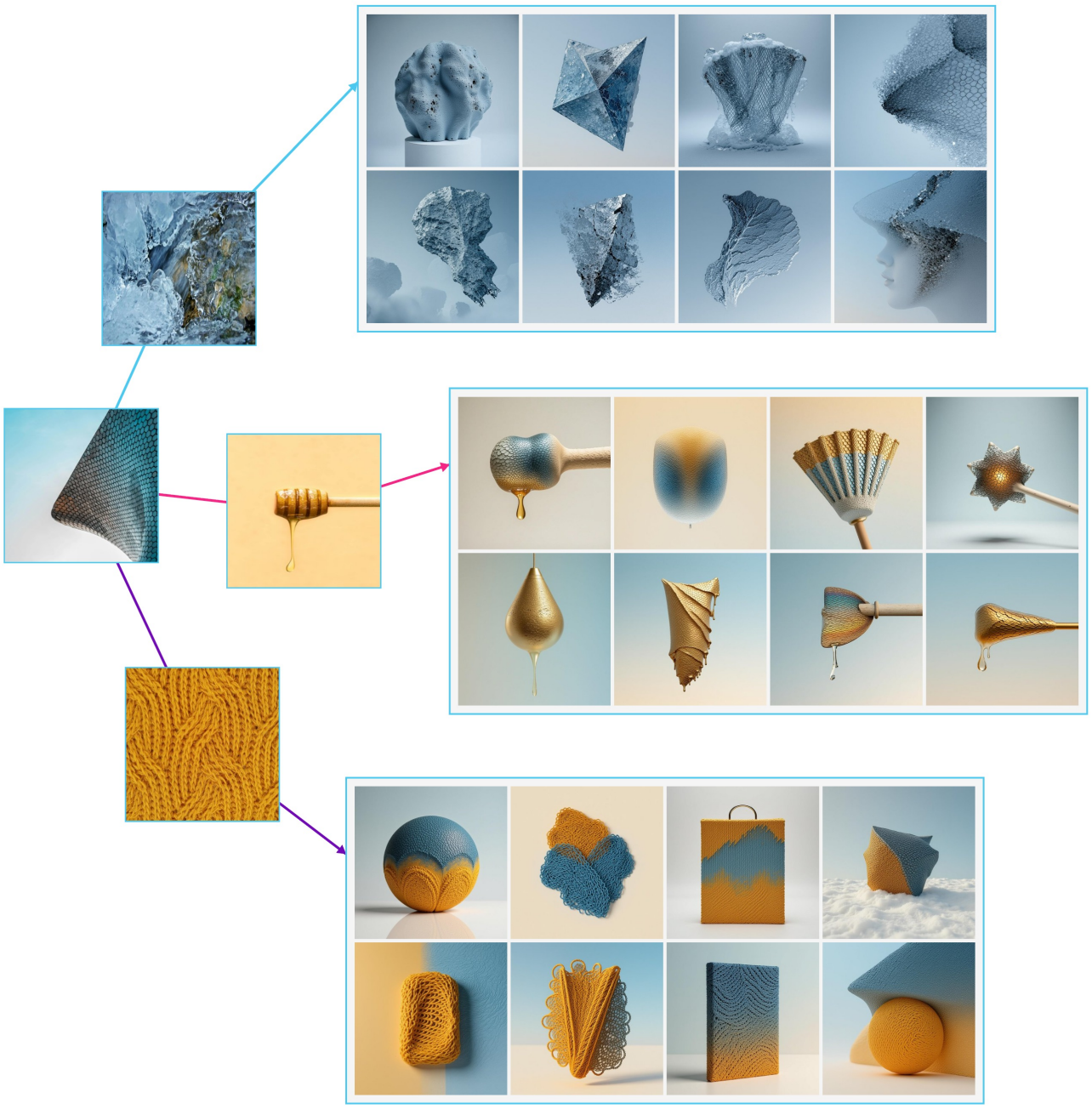


Fig. 17. Exploration canvas showing visual combinations generated by our method.

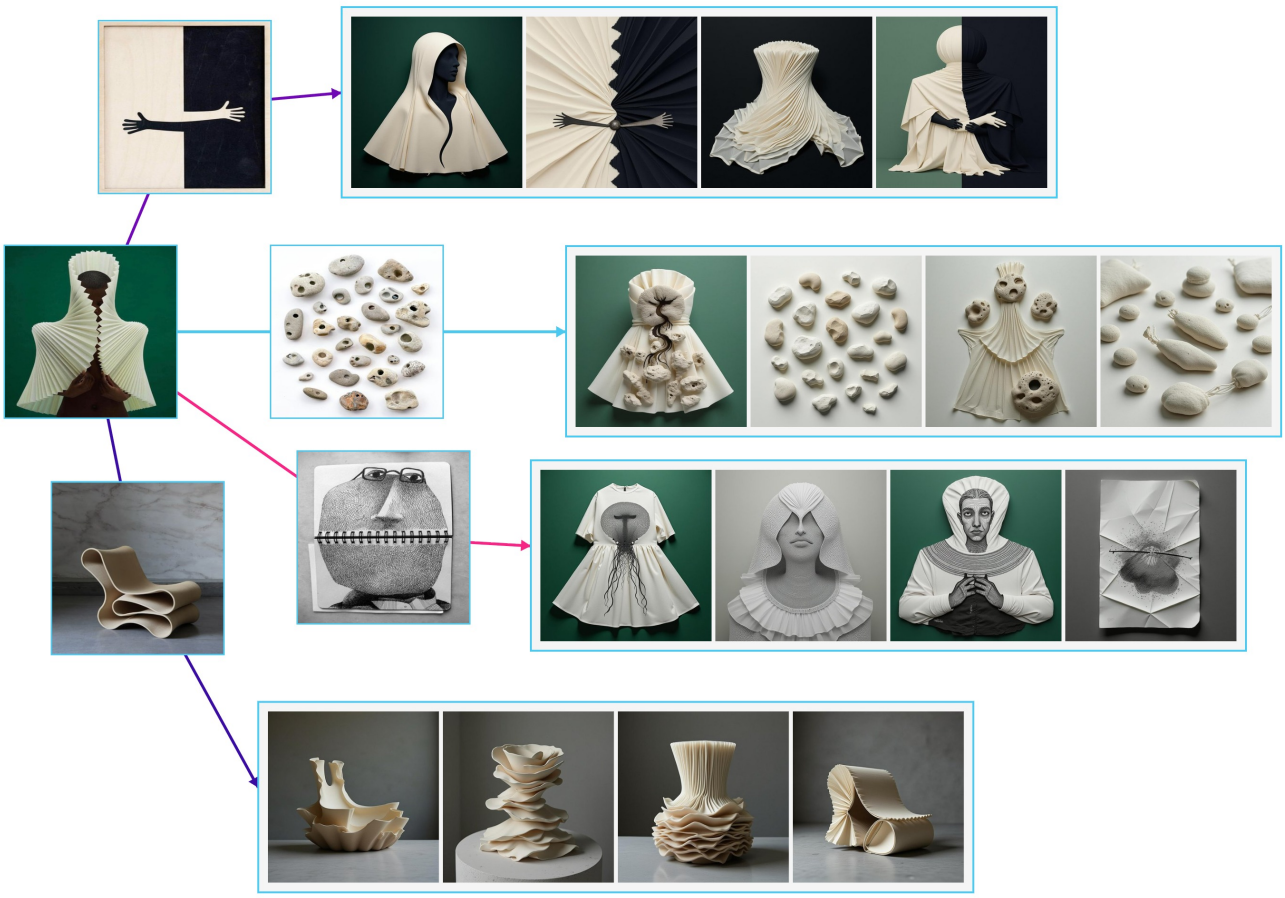


Fig. 18. Exploration canvas showing visual combinations generated by our method.

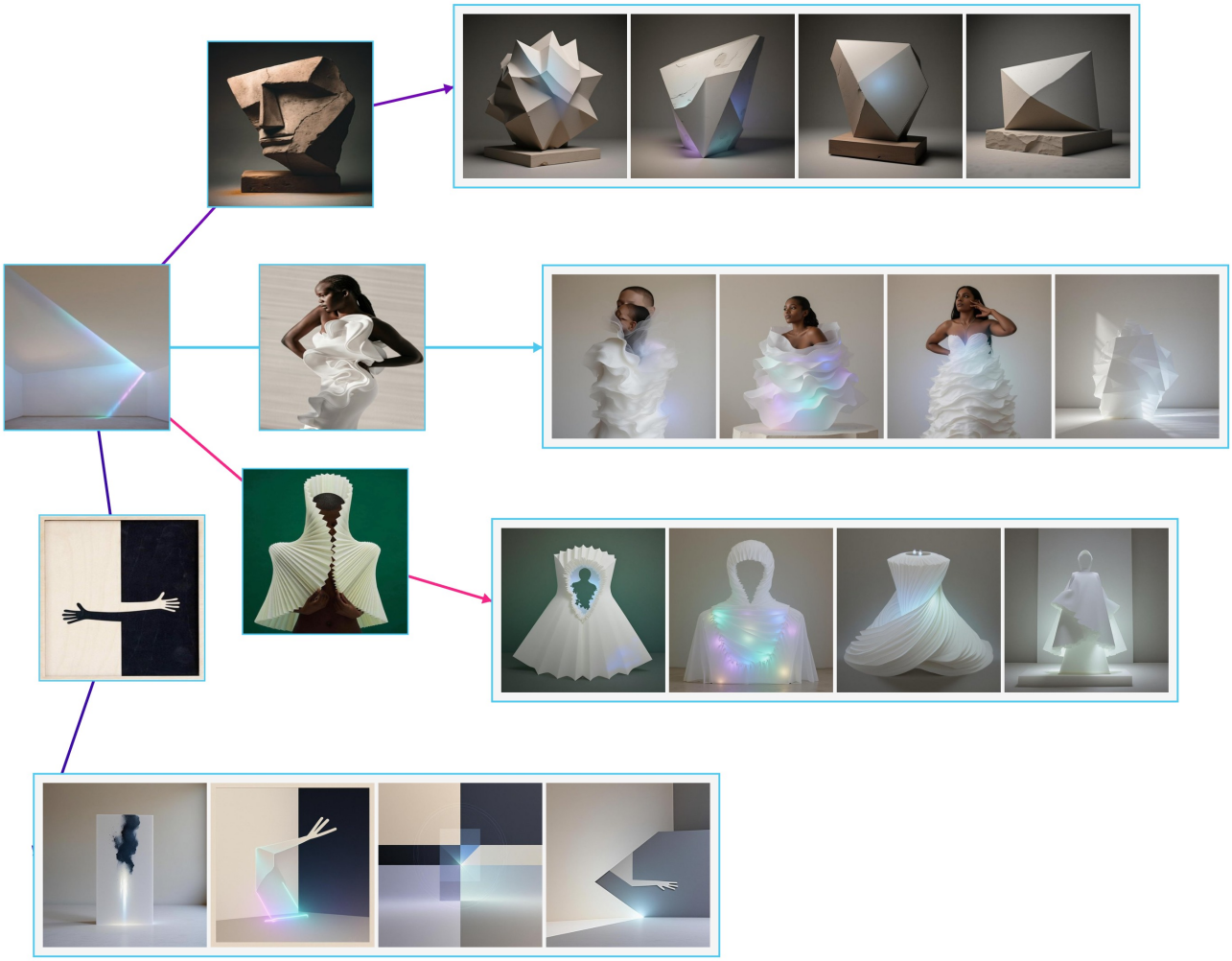


Fig. 19. Exploration canvas showing visual combinations generated by our method.

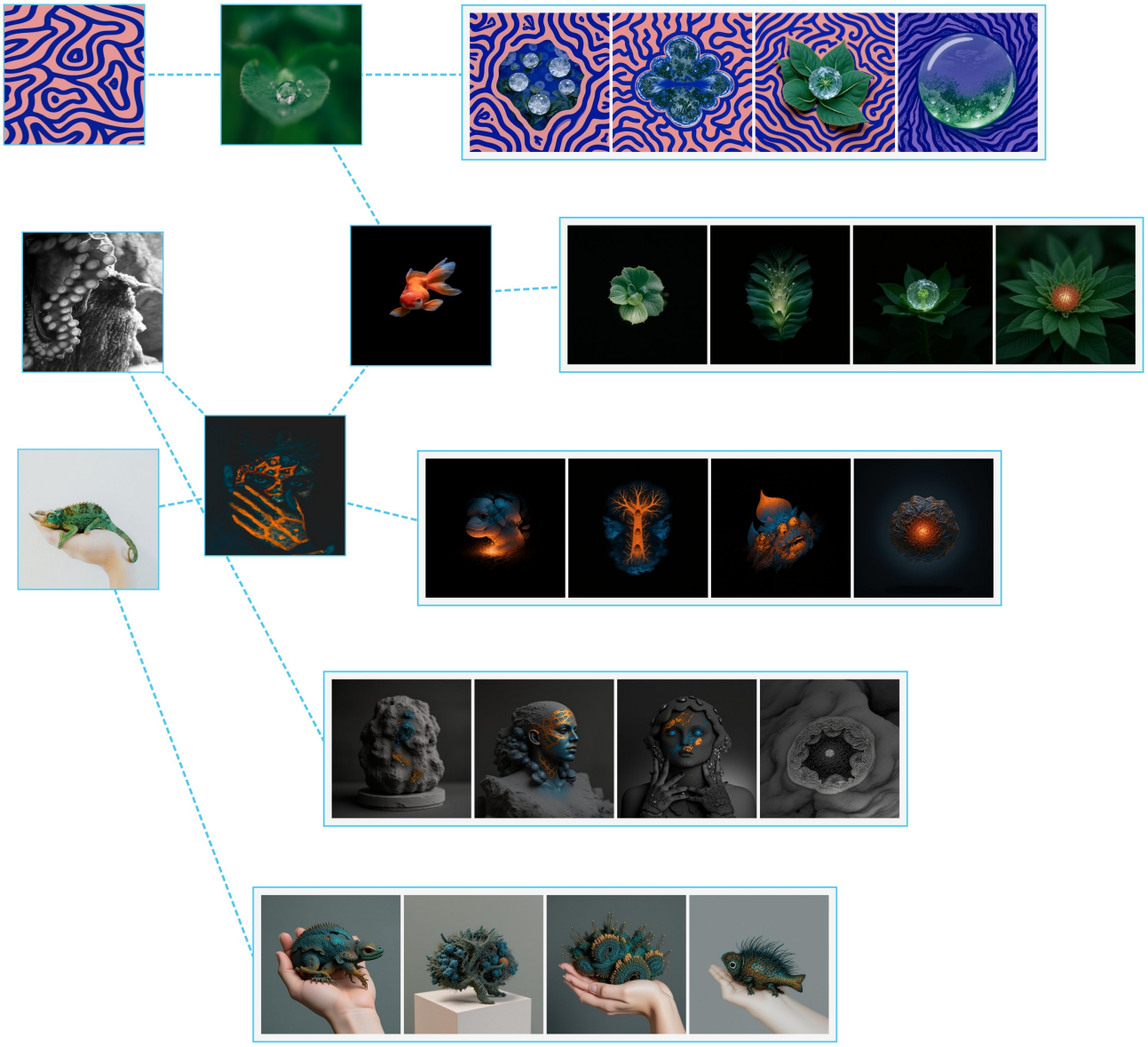


Fig. 20. Exploration canvas showing visual combinations generated by our method.

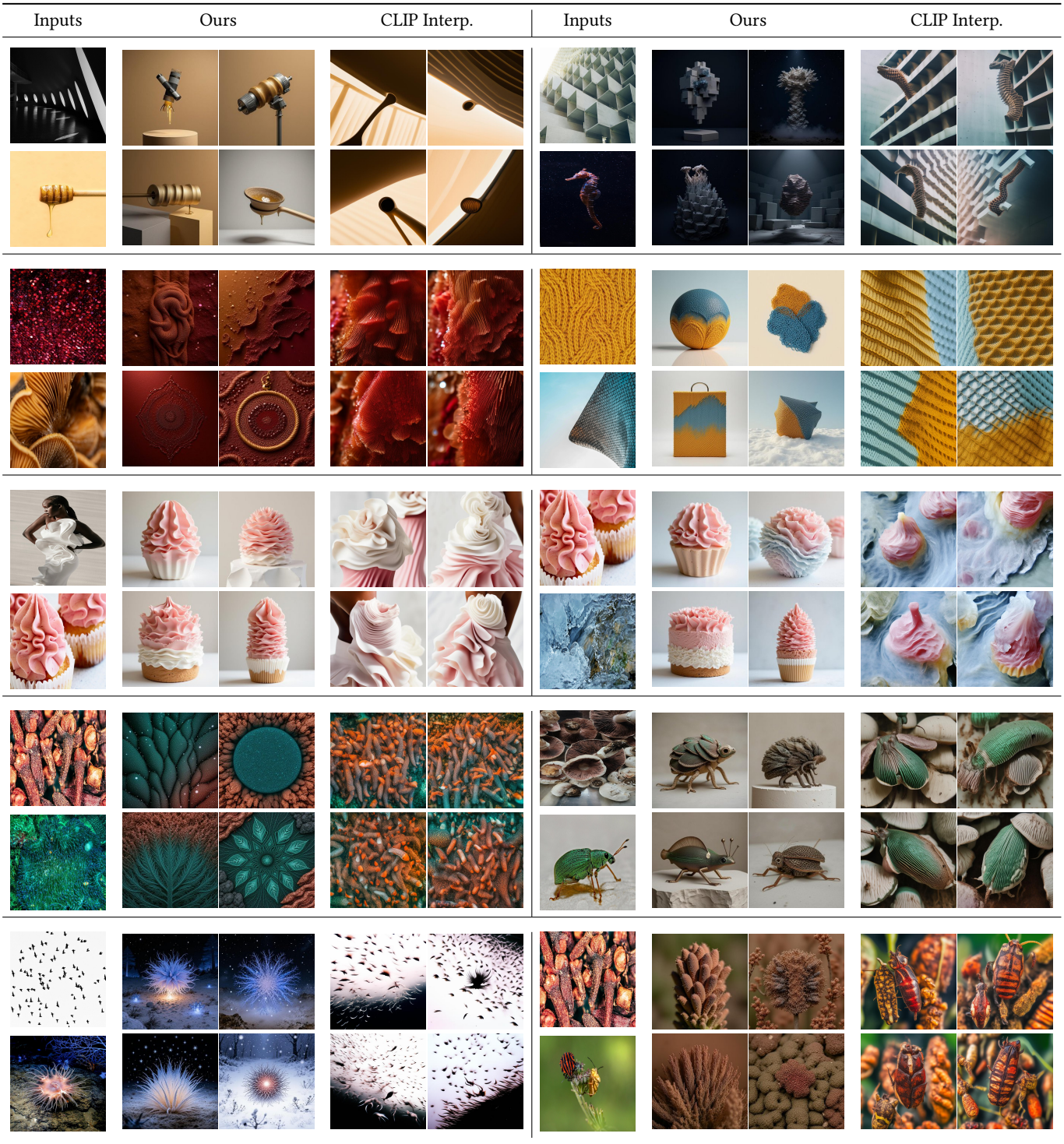


Fig. 21. Comparison with CLIP space interpolation baseline. Each row shows two input pairs, with our method's outputs (middle) and the CLIP interpolation baseline (right) for each pair. Results are shown from 4 different seeds in a 2×2 grid.

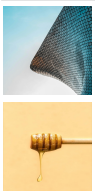
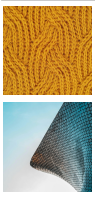
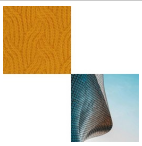
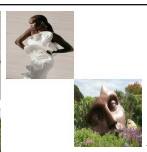
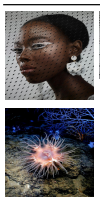
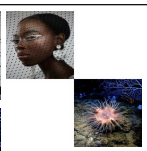
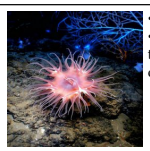
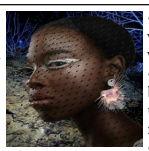
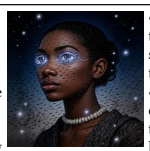
Inputs	Flux.1 Kontext	Qwen-Image-2511	Nano Banana	Ours
	 (3 words)	• copy entire grid	 (44 words)	• Use the building from image 1 as the base. • Extract the dripping honey from image 2. • Apply the extracted honey to the hexagonal surface of the building in image 1. • Adjust the honey to flow and coat the building's texture.	 (73 words)	• Use the architectural structure and hexagonal pattern from image 1. • Extract the honey color and dripping texture from image 2. • Submerge parts of the hexagonal structure from image 1 into a body of the extracted honey liquid. • Create new elements mimicking the honey drips from image 2 emerging from the liquid.	 (76 words)	• Form a fan-like object with radiating segments. • Apply the hexagonal pattern from image 1, coloring sections with blue from image 1 and golden from image 2. • For other segments, apply a ribbed texture inspired by the honey dipper's grooves (image 2) and colored light grey from image 1. • Attach the wooden handle from image 2 to the base of the object.
	 (29 words)	• Soften the knitted texture of the top-left image to a fuzzy appearance. • Emphasize the hexagonal pattern of the bottom-right image as a stark black overlay on the building's form.	 (26 words)	• Use the architectural structure and sky from image 2. • Recolor the golden-hued section of the building's facade with the yellow color from image 1.	 (43 words)	• Extract the background sky and gradient from image 2. • Extract the shape of the building from image 2. • Fill the extracted building shape with the texture from image 1. • Place the filled shape onto the background from image 2.	 (73 words)	• Create a rectangular object with a handle. • Apply the knitted texture from image 1 to the entire object. • Color the top and bottom sections of the object using the yellow from image 1. • Color the middle section using the blue/grey gradient and tones from the building in image 2. • Form the boundary between the colored sections with an angular pattern inspired by the hexagonal structure in image 2.
	 (29 words)	• Isolate the woman from the top-left image. • Integrate the isolated woman into the background of the bottom-right image. • Adjust the woman's scale and lighting to match the garden scene.	 (3 words)	• copy ⟨image2⟩	 (32 words)	• Use the scene from image2. • Extract the white ruffled material from the dress in image1. • Place the extracted ruffled material around the base of the face sculpture in image2.	 (66 words)	• Extract the sculptural head form and features from image 2. • Apply the matte, reddish-brown/terracotta color and texture from the sculpture in image 2 to the head and upper body. • Wrap the lower part of the bust with the white, ruffled, sculptural fabric texture from the dress in image 1. • Present the sculpture against a plain, light, minimalist background, similar to image 1.
	 (3 words)	• copy entire grid	 (15 words)	• Copy image 2. • Overlay the dotted texture from image 1 onto image 2.	 (47 words)	• Extract the woman's face and veil from image 1. • Replace the background with the underwater scene from image 2. • Replace the earring from image 1 with the sea anemone from image 2. • Position the sea anemone as an earring on the woman's ear.	 (86 words)	• Take the woman's face from image1. • Extract the dotted pattern from the veil in image1, stylize these dots into arrow shapes, and overlay them on the face and in the background. • Enhance the white eyeliner from image1 to create a glowing effect for the eyes. • Integrate the translucent, tentacle-like structures and luminosity from the sea anemone in image2 into the glowing details of the eyes. • Adopt the dark, atmospheric color palette and subtle light sources from image2 for the overall background.

Fig. 22. Description complexity comparison (1/2). Each row shows two input images and outputs from four methods. Below each output, we show the VLM-generated instructions describing how to recreate it, along with the word count. Our method produces outputs that require significantly more words to describe, indicating more complex and non-trivial combinations.

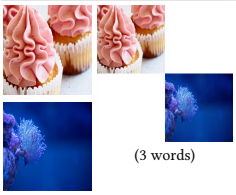
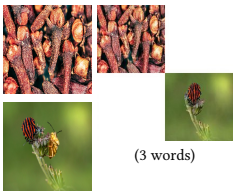
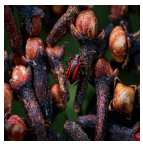
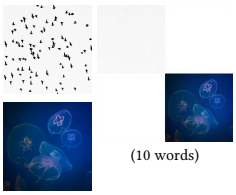
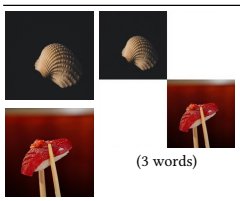
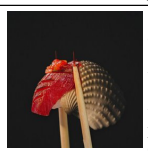
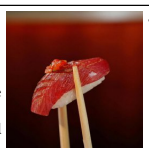
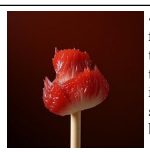
Inputs	Flux.1 Kontext	Qwen-Image-2511	Nano Banana	Ours
	• copy entire grid (3 words)	 • Create a vertically split image. • Place image2 on the left side. • Place scaled-down cupcakes from image1 onto a white speckled counter on the right side. (28 words)	 • Place the cupcakes from image1 into the underwater scene from image2. • Apply the intricate texture and shape of the coral/anemone from image2 to the pink frosting of the cupcakes. • Incorporate other coral elements from image2 into the foreground and background of the scene. (46 words)	 • Use the texture and color of the frosting from ⟨image1⟩ to create pink flower petals and ruffled layers. • Extract the tentacle texture and overall form of the coral from ⟨image2⟩, recoloring it purple, for the fluffy middle layer. • Apply the texture of the frosting from ⟨image1⟩ to create a white ruffled layer. • Extract the color and textured surface from the cupcake base of ⟨image1⟩ for the bottom layer. • Use the gradient background style from ⟨image2⟩, recolored to purple. (83 words)
	• copy bottom-right image (3 words)	 • Place the striped bug from image 2 onto the cloves from image 1. • Replace the background of the clove composition with a blurred green, inspired by image 2. (30 words)	 • Isolate the bugs and the plant stem from image 2. • Use the cloves from image 1 as the new foreground/surface. • Place the isolated bugs and plant stem onto the cloves, retaining the blurred green background from image 2. (41 words)	 • Extract and multiply individual clove buds and stalks from Image 1. • Arrange these elements to form a dense spherical structure for the main object and smaller clusters for background details. • Apply a strong desaturation and color shift to achieve a muted brown/tan palette. • Integrate darker, hue-shifted clove elements from Image 1 into the main sphere for visual accents. • Apply a shallow depth of field blur to the background and surrounding elements, similar to the effect in Image 2. (83 words)
	• Replace the birds image in the top-left quadrant with white. (10 words)	 • Extract jellyfish from image2. • Convert extracted jellyfish to grayscale outlines/sketches. • Place the outlined jellyfish on a white background. • Extract birds from image1. • Overlay birds from image1 onto and around the jellyfish. (36 words)	 • Copy image 2. • Extract birds from image 1. • Place and scale down multiple groups of extracted birds onto the jellyfish in image 2. (26 words)	 • Establish a deep blue background and luminous glow, inspired by the environment of Image 2. • Create a translucent spherical object as the central focal point. • Generate intricate floral patterns by abstracting and stylizing the internal structures of the jellyfish from Image 2. • Apply the sharp definition and silhouette-like quality from Image 1 to these patterns, rendering them with the glowing blue color and translucency of Image 2. • Add blue, glowing cloud formations at the base, consistent with the ethereal atmosphere of Image 2. (88 words)
	• copy entire grid (3 words)	 • Extract the sushi and chopsticks from ⟨image2⟩. • Extract the seashell from ⟨image1⟩. • Combine these two elements, placing the sushi and chopsticks in front of the seashell. • Use the dark background from ⟨image1⟩. (36 words)	 • copy ⟨image2⟩ (3 words)	 • Use the seashell object and its intricate texture from image 1. • Apply the red color and glossy, translucent material from the tuna in image 2 to the seashell shape. • Extract one chopstick from image 2 and position it to hold the modified seashell. • Place the resulting object in the background from image 2. (57 words)

Fig. 23. Description complexity comparison (2/2). Continued from previous figure. Note how Kontext consistently produces grid layouts, while Qwen and Nano Banana sometimes simply copy one input. Our method consistently generates non-trivial combinations requiring detailed descriptions.

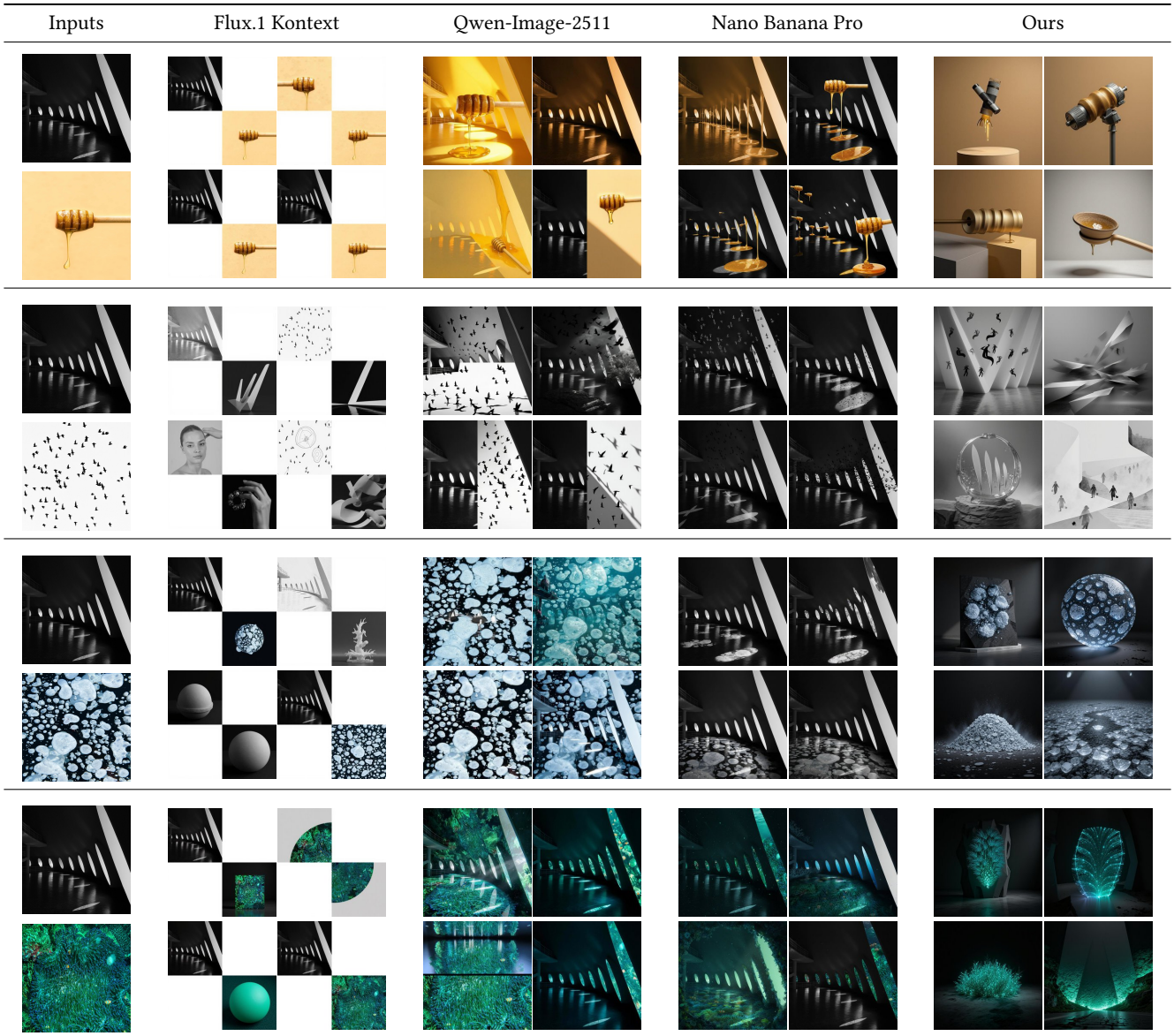


Fig. 24. Generation results comparison (1/5). Each row shows two input images (left) and outputs from four methods, each displaying results from 4 different seeds in a 2×2 grid.

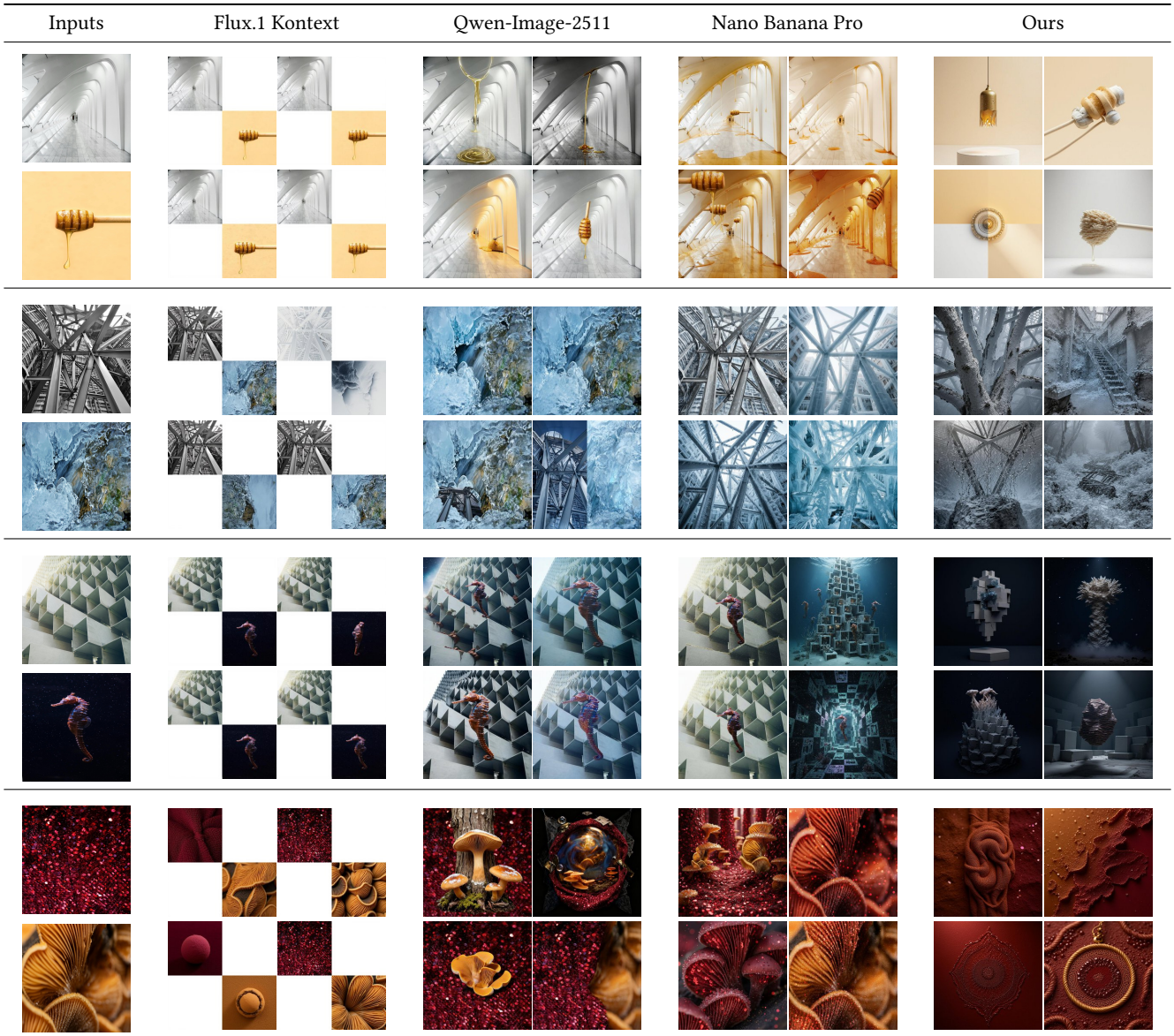


Fig. 25. Generation results comparison (2/5). Continued from previous figure.

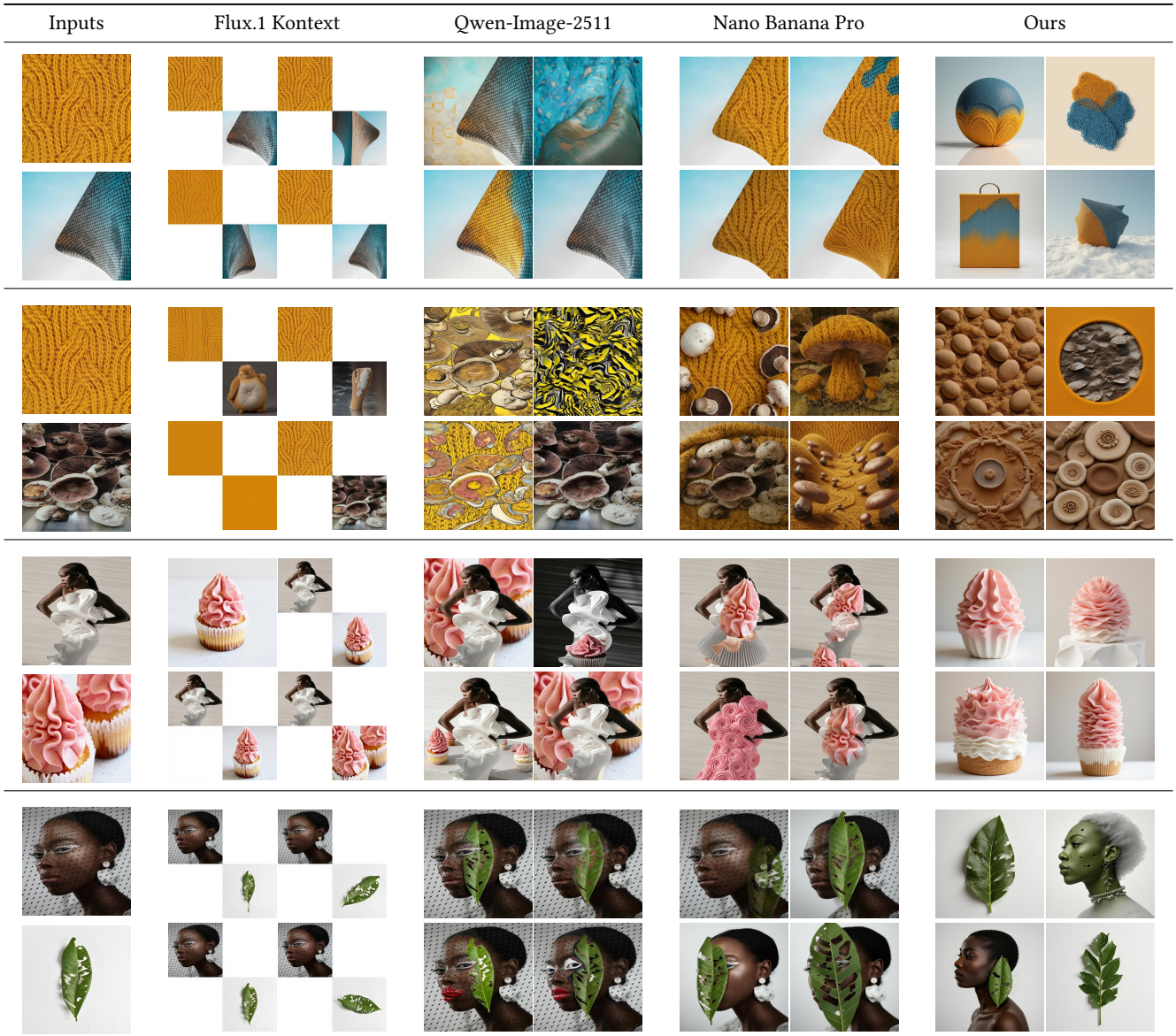


Fig. 26. Generation results comparison (3/5). Continued from previous figure.

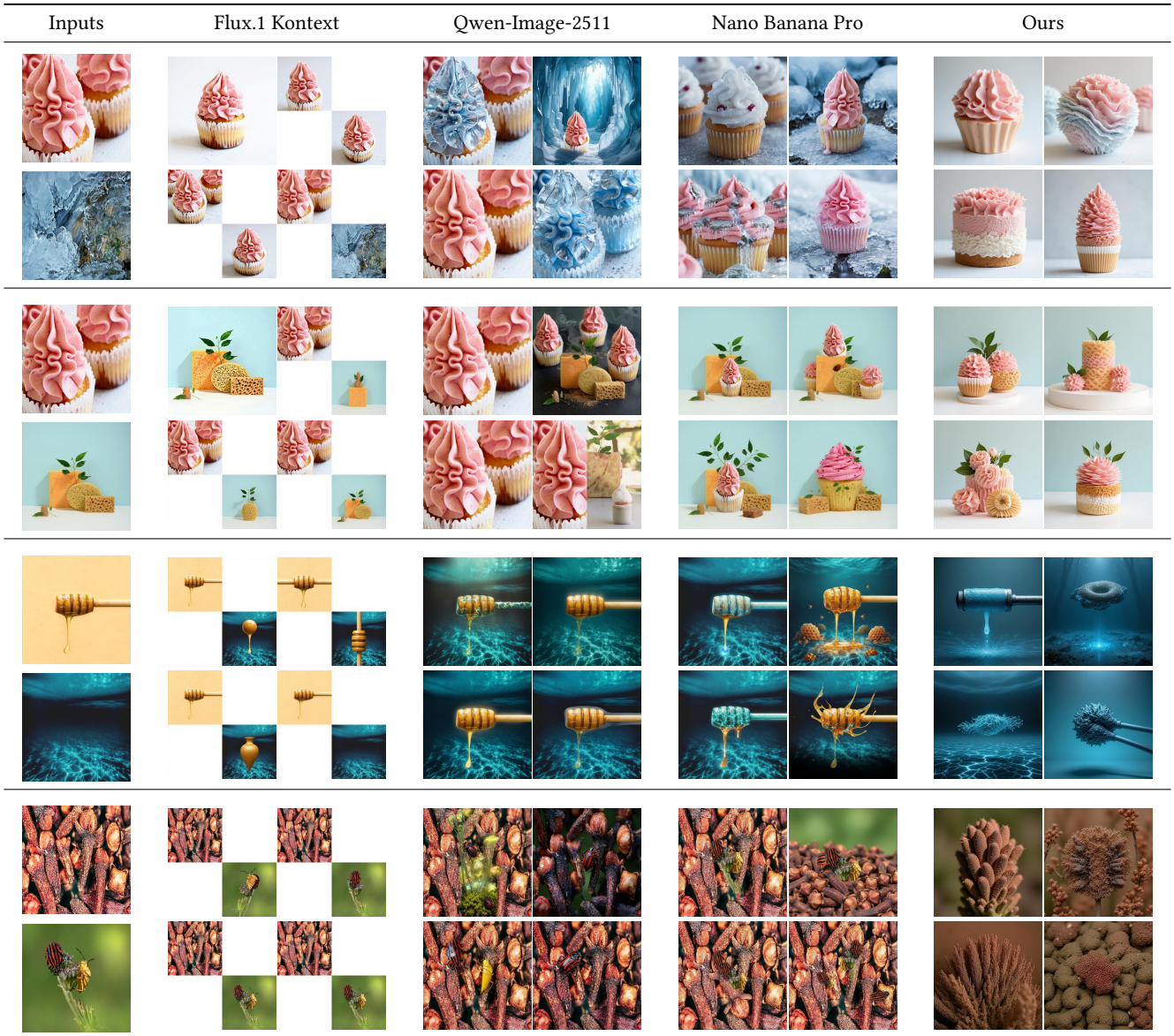


Fig. 27. Generation results comparison (4/5). Continued from previous figure.

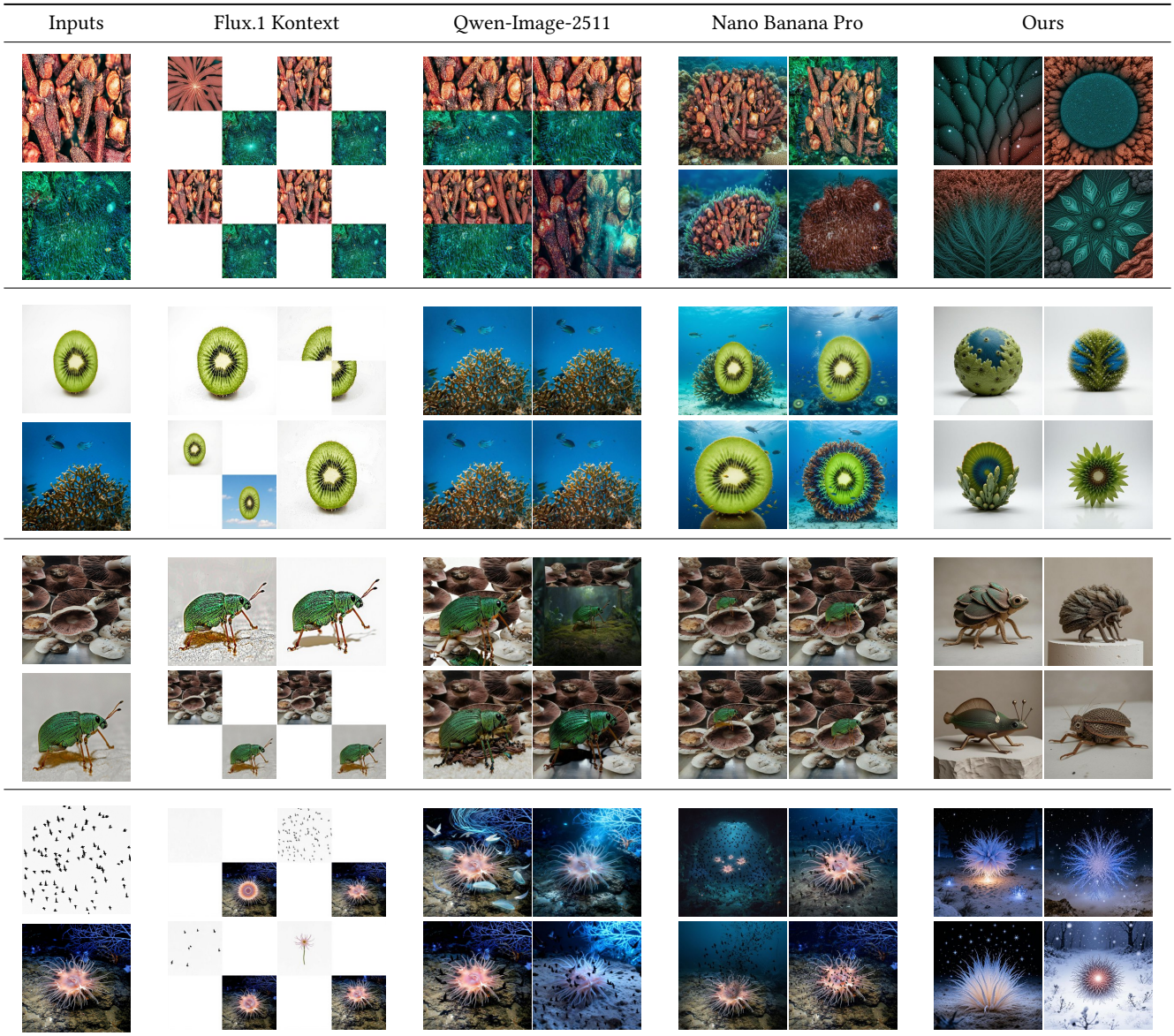


Fig. 28. Generation results comparison (5/5). Continued from previous figure.

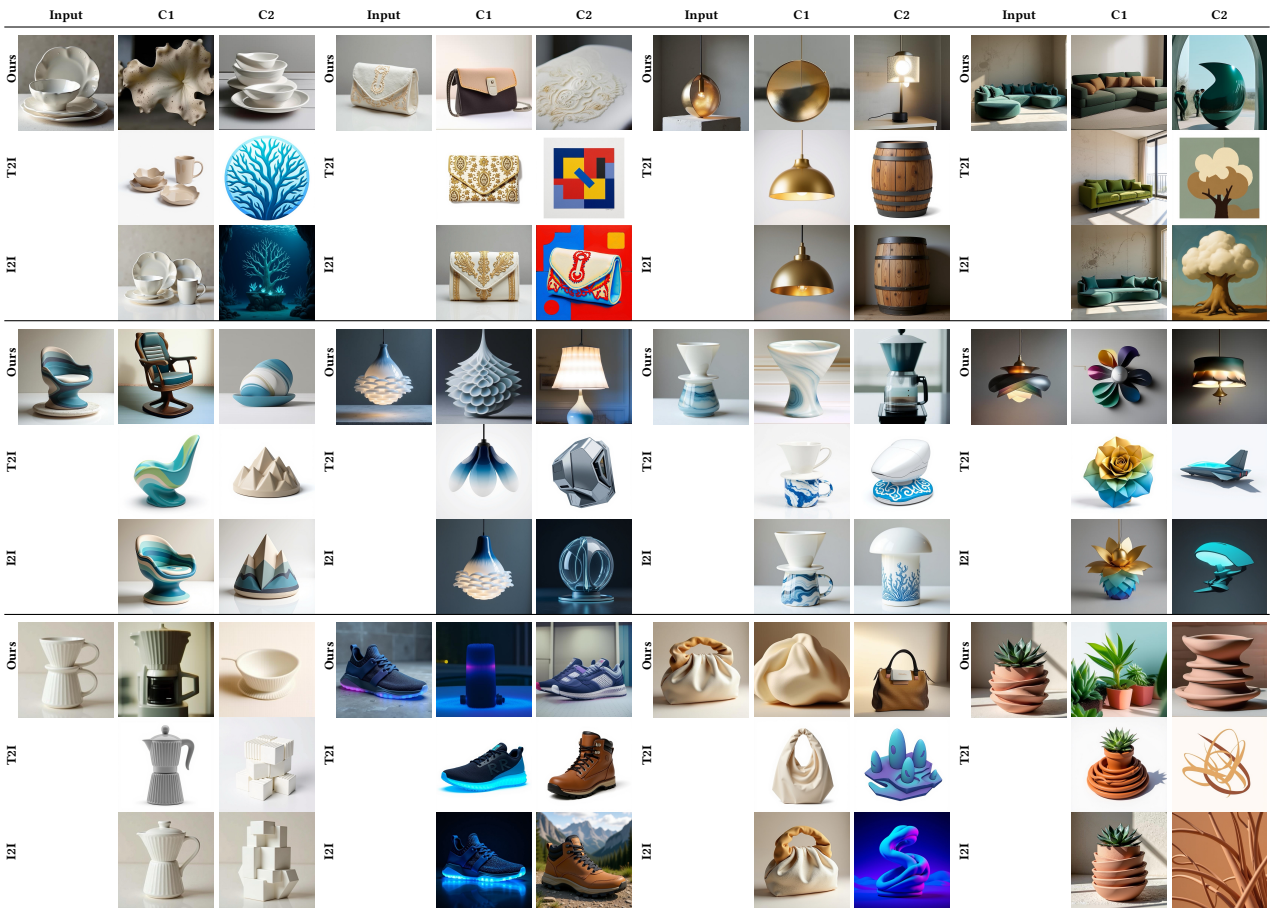


Fig. 29. Decomposition results comparison. Given an input image, we decompose it into two components (C1, C2) using three methods. Our SAE-based approach produces components that capture distinct visual aspects while maintaining semantic relevance. The T2I baseline generates from VLM-provided text prompts only, while I2I uses the input image with text prompts.